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Chapter 1
Coordinate calculus

Abstract We review techniques to do computations by tensors. Coordinate calculus means that we encode information in the coordinates of the tensors. As a major example we investigate generic Fourier transforms.
Fourier transform as an example.

1.1 Group fields

A group field is a vector

$$\tau\left[X\right] \,:\, G \to \mathbf{S}$$

where X enumerates the $|G| = m$ elements of the group, such that the x-th element of the group is mapped to $\tau\left[X = x\right]$. Following the generalization of contractions, $\mathbf{S}$ is a semiring. A complex group field is a group field for the complex sum-product semiring.

Example 1.1 Unfolding is not an isomorphism Let us note that the unfolding of $[\prod_{k\in[d]} m_k]$ to $\bigtimes_{k\in[d]}[m_k]$ is not a group isomorphism. The group operation in $\prod_{k\in[d]} m_k$ is the integer summation

$$x \circ y = x + y \mod \left(\prod_{k\in[d]} m_k\right).$$

In the unfolding the group operation is

$$(x_0,\ldots,x_{d-1}) \circ (y_0,\ldots,y_{d-1}) = (x_0 + y_0 \mod m_0,\ldots,x_{d-1} + y_{d-1} \mod m_{d-1}).$$

Intuitively, the difference of these groups is that in the unfolded group summation carry bits are set to 0.

The group field perspective on tensors further applied in quantum gravity. Here symmetries are modeled based on the group structure [5, 4].

1.1.1 Coset encoding

We here restrict to finite abelian groups. A subgroup is a subset $H \subset G$, which is itself a group. A coset to H and an element $g \in G$ is the set

$$\{g \circ h \,:\, h \in H\}.$$

The dual group $\hat{G}$ is the collection of all characters of irreducible representations

$$\chi_k \,:\, G \to \mathbb{C}\,,$$

with the group operation $\chi_{k_0} \circ \chi_{k_1}(g) = \chi_{k_0}(g) \cdot \chi_{k_1}(g)$. The annihilator of a subgroup $H \subset G$ is the subgroup

$$H^\perp := \{\chi_k \,:\, \chi_k \in \hat{G},\, \forall h \in H : \chi_k\,(h) = 1\}\,.$$

The annihilator of the annihilator is

$$\left(H^\perp\right)^\perp = H\,.$$

Since finite Abelian group are isomorphic to cartesian products of cyclic groups, we assume $G = \bigtimes_{k\in[d]} \mathbb{Z}_{m_k}$. The characters are enumerated by $z_{[d]} \in \bigtimes_{k\in[d]} \mathbb{Z}_{m_k}$ and defined as the maps

$$\chi_{z_{[d]}}(x_{[d]}) = \exp\left[2\pi \mathrm{i} \sum_{k\in[d]} \frac{z_k \cdot x_k}{m_k}\right]\,.$$

The annihilator of a subgroup generated by $h^j_{[d]}$ for $j \in [n]$ is thus defined by

$$\{\chi_{z_{[d]}} \,:\, \forall j \in [n] : \sum_{k\in[d]} \frac{z_k \cdot h^j_k}{m_k} = 0 \mod 1\}\,.$$

We build for each $h^j_{[d]}$ an edge

$$e_j = \{k \,:\, h^j_k \neq 0\}$$

and a tensor

$$\tau^{h^j_{[d]}}\left[X_{e_j} = x_{e_j}\right] = \begin{cases} \mathbf{1} & \text{if} \quad \sum_{k\in e_j} \frac{z_k \cdot h^j_k}{m_k} = 0 \mod 1 \\ \mathbf{0} & \text{else} \end{cases}\,.$$

Then we have for the subset encoding of $H^\perp \subset \hat{G}$ (where $\hat{G}$ is enumerated by $x_{[d]} \in \bigtimes_{k\in[d]} [m_k]$) that

$$\epsilon_{H^\perp}\left[X_{[d]}\right] = \left\langle \{\tau^{h^j_{[d]}}\left[X_{e_j} = x_{e_j}\right] \,:\, j \in [n]\}\right\rangle_{\left[X_{[d]}\right]}\,.$$

Example 1.2 If $m_k = 2$ we can simplify the support condition for $\tau^{h^j_{[d]}}\left[X_{e_j} = x_{e_j}\right]$, since any nonvanishing h^j_k is 1. The condition becomes

$$\sum_{k\in e_j} z_k = 0 \mod 2$$

Notice, that this is captured by the generalized exactly or connective and thus

$$\tau^{h^j_{[d]}}\left[X_{e_j}\right] = \bigoplus [X_{e_j}]\,.$$

Using the associativity of the exactly or connective, one can decompose the tensor in an arbitrary tree format with hidden ranks 2.

Having represented a group itself based on its annihilator, we have represented the coset of H to the neutral group element 0. To represent an arbitrary coset to an element $z_{[d]}$ we apply a shift operator

$$\bigotimes_{k\in[d]} \tau^k\left[X_k, \tilde{X}_k\right]\,,$$

where each local shift operator is a basis encoding of a shift function defined as

$$\tau^k \left[X_k = x_k, \tilde{X}_k = \tilde{x}_k\right] = \begin{cases} 1 & \text{if} \quad x_k + z_k = \tilde{x}_k \mod m_k \\ 0 & \text{else} \end{cases} .$$

1.2 Fourier transform

Fourier transforms are based on matrix representations of groups, and therefore defined based on the complex sum-product semiring.

1.2.1 Unfolding

The Fourier transform of vectors in $\mathbb{C}^m$ is represented by a matrix

$$\mathcal{FT}\left[X = x, Y = y\right] = \frac{1}{\sqrt{m}} \exp\left[2\pi \mathrm{i} \cdot \frac{x \cdot y}{m}\right]$$

where X, Y are variables of dimension m. The Fourier transform of a vector $V[X]$ is then the contraction

$$\langle V[X], \mathcal{FT}\left[X, Y\right] \rangle_{[Y]} .$$

We assume $m = 2^d$ and represent each index $x \in [2^d]$ by a tuple $(x_0, \ldots, x_{d-1}) \in [2]^d$ such that

$$x = \sum_{k \in [d]} x_k \cdot 2^{d-k-1} .$$

Note that in this convention, x_{d-1} is the least significant bit. We follow [22] and denote x by $x_0 \ldots x_{d-1}$. By $x_0 \ldots x_{k-1}.x_k \ldots x_{d-1}$ we denote $\frac{1}{2^{d-k-1}} \cdot x$.

The unfolded vector $V[X]$ is a tensor $V[X_{[d]}]$ with indices

$$V[X_{[d]} = x_{[d]}] = V[X = \left(\sum_{k \in [d]} x_k \cdot 2^{d-k-1}\right)] .$$

In the unfolded representation the fourier transform is described as the contraction with a tensor $\mathcal{FT}^d[X_{[d]}, Y_{[d]}]$ with coordinates

$$\mathcal{FT}^d[X_{[d]} = x_{[d]}, Y_{[d]} = y_{[d]}] = \frac{1}{\sqrt{2^d}} \exp\left[2\pi \mathrm{i} \cdot \left(\sum_{k \in [d]} x_k \cdot 2^{d-k-1}\right) \cdot \left(\sum_{k \in [d]} y_k \cdot 2^{d-k-1}\right)\right] ,$$

that is the Fourier transform of $V[X_{[d]}]$ is

$$\left\langle V[X_{[d]}], \mathcal{FT}\left[X_{[d]}, Y_{[d]}\right] \right\rangle_{[Y_{[d]}]} .$$

1.2.2 Tensor network decomposition

We now derive a tensor network decomposition of the tensor $\mathcal{FT}^d[X_{[d]}, Y_{[d]}]$.

Lemma 1.1 *The Fourier transform of basis vectors $\epsilon_{x_{[d]}}\left[X_{[d]}\right]$ is the elementary tensor*

$$\frac{1}{\sqrt{2^d}} \bigotimes_{k\in[d]} \left(\epsilon_0\left[Y_k\right] + \exp\left[2\pi\mathrm{i}\cdot(0.x_{d-k-1}\cdots x_{d-1})\right]\cdot\epsilon_1\left[Y_k\right]\right)$$

Proof. For each $y_{[d]}$ we have

$$\begin{aligned}
&\left\langle \{(\epsilon_0\left[Y_k\right] + \exp\left[2\pi\mathrm{i}\cdot 0.x_{d-k-1}\cdots x_{d-1}\right]\cdot\epsilon_1\left[Y_k\right]) \,:\, k\in[d]\} \cup \{\epsilon_{y_{[d]}}\left[Y_{[d]}\right]\}\right\rangle_{[\varnothing]} \\
&= \prod_{k\in[d]} \exp\left[2\pi\mathrm{i}\cdot(0.x_{d-k-1}\cdots x_{d-1})\cdot y_k\right] \\
&= \prod_{k\in[d]} \exp\left[2\pi\mathrm{i}\cdot(x_0\cdots x_{d-k-2}.x_{d-k-1}\cdots x_{d-1})\cdot y_k\right] \\
&= \exp\left[2\pi\mathrm{i}\cdot x\sum_{k\in[d]} y_k\cdot 2^{d-k-1}\right] \\
&= \exp\left[2\pi\mathrm{i}\cdot\frac{x\cdot y}{2^d}\right].
\end{aligned}$$

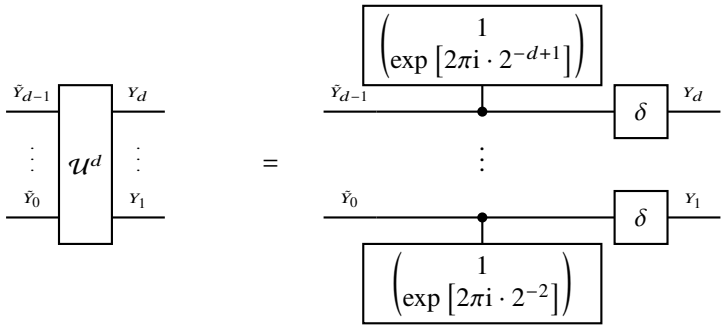
Since the basis coefficients of the elementary tensor coincide with those of the Fourier transform of $\epsilon_{x_{[d]}}\left[X_{[d]}\right]$, both tensors are equal. □

We introduce the diagonal unitary

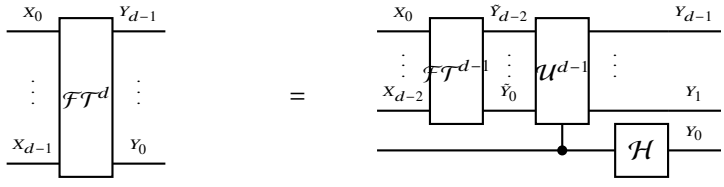
$$\mathcal{U}^d[X_{[d]} = x_{[d]}, Y_{[d]} = y_{[d]}] = \begin{cases} \exp\left[2\pi\mathrm{i}\cdot\left(\sum_{k\in[d]} x_k\cdot 2^{-k-2}\right)\right] & \text{if} \quad x_{[d]} = y_{[d]} \\ 0 & \text{else} \end{cases}$$

Note that $\mathcal{U}^d$ is diagonal and the tensor product of matrices with variables X_k, Y_k. Diagonality can be used to contract $\mathcal{U}^d$ to an arbitrary tensor with computational demand in $\curlywedge(2^d)$.

Using diagonality and elementarity, we have



Lemma 1.2 *The Fourier transform on* $[2^d]$ *can be decomposed into a Fourier transform on* $[2^{(d-1)}]$ *as:*



Proof. It suffices to show that the Fourier transform of arbitrary basis tensors $\epsilon_{x_{[d]}}\left[X_{[d]}\right]$ follows the decomposition. The Fourier transform of $\epsilon_{[d-1]}\left[X_{[d-1]}\right]$ is by the Lemma above

$$\frac{1}{\sqrt{2^{d-1}}} \bigotimes_{k\in[d-1]} \left(\epsilon_0\left[\tilde{Y}_k\right] + \exp\left[2\pi\mathrm{i}\cdot(0.x_{d-k-2}\cdots x_{d-2})\right]\cdot\epsilon_1\left[\tilde{Y}_k\right]\right)$$

In case of $x_{d-1} = 0$, the Fourier transform of $\epsilon_{x_{[d]}}\left[X_{[d]}\right]$ is the tensor product of this state with $\epsilon^+\left[X_0\right]$. This is reproduced by the decomposition, since the controlled unitary $\mathcal{U}^{d-1}$ acts trivially and the Hadamard gate transforms $\epsilon_0\left[d-1\right]$ to $\epsilon^+\left[X_0\right]$.

In case of $x_{d-1} = 1$, the leg vectors $\left(\epsilon_0\left[\tilde{Y}_k\right] + \exp\left[2\pi\mathrm{i}\cdot(0.x_{d-k-2}\cdots x_{d-2})\right]\cdot\epsilon_1\left[\tilde{Y}_k\right]\right)$ are further transformed into

$$\left(\epsilon_0\left[Y_{k+1}\right] + \exp\left[2\pi\mathrm{i}\cdot(0.x_{d-k-2}\cdots x_{d-2}x_{d-1})\right]\cdot\epsilon_1\left[Y_{k+1}\right]\right)\ .$$

In both cases we arrive at the Fourier transform of $\epsilon_{x_{[d]}}\left[X_{[d]}\right]$. □

By iteratively decomposing $\mathcal{FT}^{d-1}$ until we arrive at $\mathcal{FT}^1$ (which is the $\mathcal{H}$ matrix), we get a decomposition of $\mathcal{FT}^d$ into d $\mathcal{H}$ matrices and controlled U unitaries.

1.2.3 Fast Fourier Transform

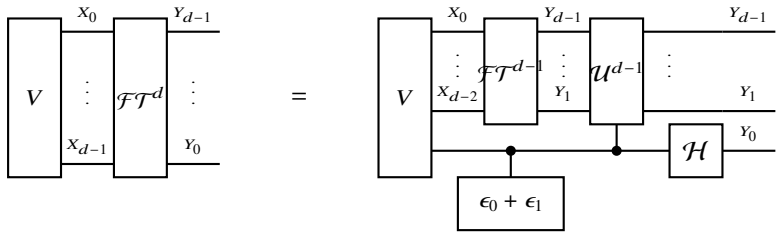
The FFT has been studied in the tensor formalism, see [9] and Chapter 14 in [14].

1.2.3.1 Layerwise contraction

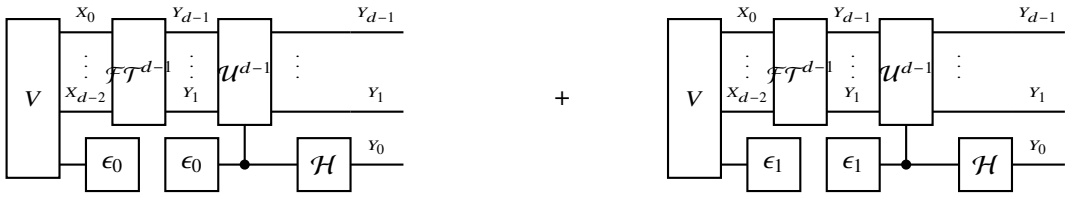
We contract the FFT tensor network layerwise. There are $2\cdot d$ layers, each has a contraction demand in $O(2^d)$ (exploiting diagonality of U). The algorithm has a total computational demand in $O(d2^d)$.

1.2.3.2 Divide and conquer

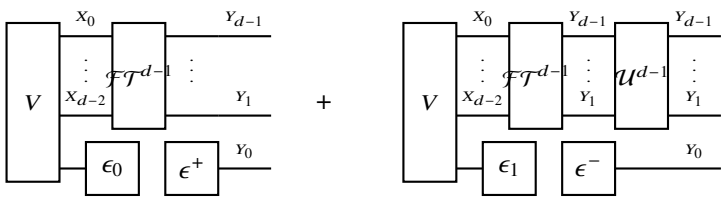
Called the radix-2 FFT [7]. Can be seen as an iterative simplification by adding a $\mathbb{I}\left[X_{d-1}\right]$ and splitting it into the basis vectors.



Now by linearity of contractions, this splits into two terms



We simplify them to



The contraction of V with $\epsilon_1\left[X_{d-1}\right]$ respectively $\epsilon_1\left[X_{d-1}\right]$ are the even and the odd coordinates of V. Both are Fourier transformed with $d-1$ bits, which are further divided into Fourier transforms of lower order. In the conquer step of the FFT, the odd fourier transform gets a coordinatewise transform by U, which is efficient since U is diagonal.

1.2.4 Generic groups

So far we have treated the Fourier transform on the cyclic group $\mathbb{Z}_{\prod_{k\in[d]} m_k}$ using unfoldings. More naturally, a tensor is defined as a complex field on the cartesian product group $\bigtimes_{k\in[d]}[m_k]$. Let us now generalize to generic groups.

1.2.4.1 Representations

Recall that tensors are group fields, see Sect. 1.1.

A representation is a map of G to a matrix group. Irreducible representations (irrep) are those, which are not equivalent to matrix groups in block form. We enumerate them by a variable Y and store all irreps in the tensor

$$\rho^G\left[X, Y, L^{\text{in}}, L^{\text{out}}\right] ,$$

where the y-th representation maps the group element x-th to the matrix

$$\frac{1}{\sqrt{p}} \cdot \rho^G\left[X = x, Y = y, L^{\text{in}}, L^{\text{out}}\right] .$$

Here p is the dimension of the representation. We here made the convention to store in the slices of ρ^G the by $\sqrt{p}$ rescaled irrep, to ease the definition of the Fourier transform.

Example 1.3 Cyclic group For the cyclic group $G = \mathbb{Z}_m$ the irreps are one-dimensional, that is $n = 1$, and enumerated by $y \in [m]$ as

$$\rho^{\mathbb{Z}_m}\left[X = x, Y = y, L^{\text{in}} = 0, L^{\text{out}} = 0\right] = \exp\left[2\pi\mathrm{i} \cdot \frac{x \cdot y}{m}\right] .$$

The characters of the irreps are

$$\chi_y\left(x\right) = \rho^{\mathbb{Z}_m}\left[X = x, Y = y, L^{\text{in}} = 0, L^{\text{out}} = 0\right] .$$

Note that for $m = 2$ this is the Hadamard matrix

$$\rho^{\mathbb{Z}_2}\left[X, Y, L^{\text{in}} = 0, L^{\text{out}} = 0\right] = \sqrt{2} \cdot \mathcal{H}\left[X, Y\right] .$$

1.2.4.2 Finite Abelian groups

Finite Abelian groups are equivalent to cartesian products of cyclic groups, that is

$$G \sim \bigtimes_{k\in[d]} \mathbb{Z}_{m_k} .$$

The irreps of finite Abelian groups are thus disconnected tensor networks

$$\rho^{\bigtimes_{k\in[d]}\mathbb{Z}_{m_k}}\left[X_{[d]}, Y_{[d]}, L_{\text{in}}^{[d]}, L_{\text{out}}^{[d]}\right] = \bigotimes_{k\in[d]} \rho^{\mathbb{Z}_{m_k}}\left[X_k, Y_k, L_{\text{in}}^k, L_{\text{out}}^k\right]$$

where each $\rho^{\mathbb{Z}_{m_k}}\left[X_k, Y_k, L_{\text{in}}^k, L_{\text{out}}^k\right]$ is the irrep tensor to $\mathbb{Z}_{m_k}$, see Example 1.3.

1.2.4.3 Fourier transform

The Fourier transform of a complex group field $\tau\left[X\right]$ is the tensor

$$\sqrt{\frac{1}{|G|}}\cdot\left\langle \tau\left[X\right],\rho^{G}\left[X,Y,L^{\text{in}},L^{\text{out}}\right]\right\rangle_{\left[Y,L^{\text{in}},L^{\text{out}}\right]}\cdot$$

Example 1.4 (Walsh-Hadamard transform) Consider the group $G=\bigtimes_{k\in[d]}\mathbb{Z}_2$, and a complex field on it by a quantum state $\psi\left[X_{[d]}\right]$. The Fourier transform is then the application of Hadamard gates on each qubit, that is

$$\left\langle \{\psi\left[X_{[d]}\right]\}\cup\bigcup_{k\in[d]}\{\mathcal{H}\left[X_k,Y_k\right]\}\right\rangle_{\left[Y_{[d]}\right]}.$$

This is the Walsh-Hadamard transform of the quantum state.

1.2.4.4 Coset transform

Let us again assume finite Abelian groups $G=\bigtimes_{k\in[d]}\mathbb{Z}_{m_k}$. Applying the Fourier transform on a coset encoding tensor gives a tensor supported at the annihilator.

Lemma 1.3 *Let $G=\bigtimes_{k\in[d]}\mathbb{Z}_{m_k}$, $H\subset G$ be a subgroup and $z_{[d]}H$ be the coset to an element $z_{[d]}$. Then*

$$\left\langle \mathcal{FT}\left[Y_{[d]},X_{[d]}\right],\epsilon_{gH}\left[X_{[d]}\right]\right\rangle_{\left[Y_{[d]}\right]}$$

is supported at the elements of $H^{\perp}$.

Proof. We have

$$\begin{aligned}\left\langle \mathcal{FT}\left[Y_{[d]},X_{[d]}\right],\epsilon_{gH}\left[X_{[d]}\right]\right\rangle_{\left[Y_{[d]}=y_{[d]}\right]} &= \frac{1}{\sqrt{|G|}}\sum_{x_{[d]}\in H}\chi_{y_{[d]}}(z_{[d]}\otimes x_{[d]})\\ &= \frac{1}{\sqrt{|G|}}\chi_{y_{[d]}}\left(z_{[d]}\right)\cdot\left(\sum_{x_{[d]}\in H}\chi_{y_{[d]}}\left(x_{[d]}\right)\right).\end{aligned}$$

Now, if $\chi_{y_{[d]}}\in H^{\perp}$, each summand is 1, and the coordinate evaluates to $\frac{|H|}{\sqrt{|G|}}\cdot\chi_{y_{[d]}}(z_{[d]})$. Using that $|H|\cdot|H^{\perp}|=|G|$, the absolute of the coordinate is

$$\sqrt{|H|}\cdot\frac{1}{\sqrt{|H^{\perp}|}}.$$

The Hilbert-Schmidt norm of the contraction restricted to $H^{\perp}$ is therefore $\sqrt{|H|}$ and coincides with the Hilbert-Schmidt norm of the coset encoding. All coordinates to $\hat{G}\backslash H^{\perp}$ have thus to vanish, which shows the claim. Alternatively one can derive the claim, based on destructive inference on the sum term. □

Example 1.5 Let us consider the group $G=\bigtimes_{k\in[d]}[2]$, which coset representing hypercores are characterized in Example 1.2. If $|H^{\perp}|=2$ we represent a coset by

$$\bigoplus[X_e]\quad\text{or}\quad\neg\bigoplus[X_e]$$

where e is the support of the generator of $H^{\perp}$. Notice that

$$\bigoplus [X_e] = \sqrt{2^d} \cdot \left(\bigotimes_{k \in e} \epsilon^+ [X_k] + \bigotimes_{k \in e} \epsilon^- [X_k] \right)$$
$$\neg \bigoplus [X_e] = \sqrt{2^d} \cdot \left(\bigotimes_{k \in e} \epsilon^+ [X_k] - \bigotimes_{k \in e} \epsilon^- [X_k] \right)$$

The Fourier transform for this group is the Walsh-Hadamard transform which turns the basis legs $\epsilon^+ [X_k]$ and $\epsilon^- [X_k]$ to $\epsilon_0 [X_k]$ respectively $\epsilon_1 [X_k]$. The Walsh-Hadamard transform therefore returns

$$\sqrt{2^d} \left(\epsilon_{0_e} [X_e] \pm \epsilon_{1_e} [X_e] \right) ,$$

which is supported at $H^\perp$.

1.2.4.5 Hidden subgroup problem

Lets assume that a function $f : G \to \mathcal{Y}$ mapping from a finite abelian group $G = \bigtimes_{k \in [d]} \mathbb{Z}_{m_k}$ to an arbitrary set $\mathcal{Y}$ is constant on the cosets of a subgroup H (called the hidden subgroup), and different on each coset. The slices

$$\beta^f \left[Y = y, X_{[d]} \right]$$

of the basis encoding to any image element enumerated by y are then coset encoding tensors and by Lem. 1.3

$$\left\langle \beta^f \left[Y = y, X_{[d]} \right], \mathcal{FT} \left[Y_{[d]}, X_{[d]} \right] \right\rangle_{\left[Y_{[d]} \right]}$$

is supported on the annihilator $H^\perp$, where each coordinate has equal support.

When retrieving the support of this transform, we can construct a generating set for $H^\perp$ and then H as $\left(H^\perp\right)^\perp$ based on the solution of a equation system in modulus calculus. This strategy is exploited in quantum computing and we will discuss it in Sect. 2.4.1.3.

Chapter 2
Quantum circuits

Abstract Quantum mechanics is formulated in tensor spaces and quantum circuits are naturally tensor networks. Quantum Circuits are naturally tensor networks. This motivates the investigation, which contractions can be efficiently performed based on quantum circuits.

2.1 Foundations

By its central axioms, quantum mechanics of multiple qubits is formulated by tensors capturing states and discrete time evolutions. Motivated by the structural similarity, we investigate how quantum circuits can be utilized for the tensor-network based approach towards efficient and explainable AI in the tnreason framework.

Quantum circuits are contractions of gate tensors and therefore tensor networks, and measurement probabilities are given by contractions.

We in this chapter investigate, how specific contractions (especially forward inference on Computation-Activation Networks) can be performed more efficiently on a Quantum Computer.

2.1.1 Quantum and Tensor Parallelism

Potential Advantage: *Quantum Parallelism* (see [29, Section 3.2.5]).

- Evaluation of multiple function values by single circuit evaluation, we will relate it with the contraction of β^f here.
- Contraction perspective: Loop-tolerant efficient contractions.
- However: Need a generic encoding scheme to exploit this advantage, which is not known yet.
- We here only provide a scheme based on post-selection, which provides a quantum advantage only through amplitude amplification. Without amplitude amplification and post-selection of samples, the encoded distribution is always uniform.
- Further challenge: Application in NISQ devises [23], instead of fault-tolerant quantum computers.

Comparison with classical side, which we can call *Tensor Network Parallelism*: Message-passing schemes for efficient contractions, but exact in limited cases.

We present the basic notation of quantum circuits using the tensor network notation used in the tnreason formalism.

2.1.2 Quantum states and quantum circuits

Quantum states are complex tensors

$$\psi\left[X_{[d]}\right] : \bigtimes_{k\in[d]} [2] \to \mathbb{C}$$

such that

$$\left\langle (\psi\left[X_{[d]}\right])^*, \psi\left[X_{[d]}\right]\right\rangle_{[\varnothing]} = \sum_{x_{[d]}\in\bigtimes_{k\in[d]}[2]} (\psi\left[X_{[d]} = x_{[d]}\right])^* \psi\left[X_{[d]} = x_{[d]}\right] = 1 \,.$$

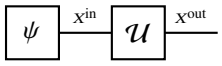
Quantum gates are unitary tensors

$$\mathcal{U}\left[X^{\text{in}}, X^{\text{out}}\right] : \bigtimes_{k\in[d]} [2] \times \bigtimes_{k\in[d]} [2] \to \mathbb{C}$$

that is

$$\left\langle (\mathcal{U}\left[X^{\text{mid}}, X^{\text{out}}\right])^\dagger, \mathcal{U}\left[X^{\text{mid}}, X^{\text{in}}\right]\right\rangle_{[=]} \delta\left[X^{\text{in}}, X^{\text{out}}\right] \,.$$

Application of quantum gates on quantum states is their contraction with respect to the complex sum-product semiring (see Sect. **??**). A quantum state $\psi\left[X^{\text{in}}\right]$ transforms under the gate $\mathcal{U}\left[X^{\text{in}}, X^{\text{out}}\right]$ to the state



.

2.1.3 Controlled single qubit gates

We define the rotation gate around the Y-axis by an angle α as

$$\mathcal{R}_Y(\alpha)\left[A^{\text{in}}, A^{\text{out}}\right] := \begin{pmatrix} \cos\left(\frac{\alpha}{2}\right) & -\sin\left(\frac{\alpha}{2}\right) \\ \sin\left(\frac{\alpha}{2}\right) & \cos\left(\frac{\alpha}{2}\right) \end{pmatrix}\left[A^{\text{in}}, A^{\text{out}}\right] \,.$$

Further we define the Pauli-X:

$$\sigma_1\left[A^{\text{in}}, A^{\text{out}}\right] := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\left[A^{\text{in}}, A^{\text{out}}\right] \,.$$

Controlled single qubit gates are defined using control qubits, where the gate is applied to the target qubit if the control qubits are in a specific state and the identity is applied otherwise. In the tensor network diagrams, we do not distinguish between incoming and outgoing control qubit variables, since the control acts as a Dirac tensor. Thus, controlled unitary with target qubit X_t and control qubits X_c are represented by tensors

$$\mathcal{U}\left[X^t_{\text{in}}, X^t_{\text{out}}, X_c\right]$$

where for each state x_c to the control variables we have that

$$\mathcal{U}\left[X^t_{\text{in}}, X^t_{\text{out}}, X_c = x_c\right]$$

is a unitary matrix acting on the leg space of the target variable.

2.1.4 Measurements and Contractions

The main difficulty of using quantum circuits as contraction providers is that we can only extract information through measurements. Therefore measurement is the only way to execute contractions of the circuit, which come with restrictions when interested in contraction with open variables.

2.1.4.1 Computational basis measurement

The computational basis measurement of the qubits X_A of a Quantum State $\psi\left[X_{[d]}\right]$ is equal to drawing samples from a distribution

$$\mathbb{P}\left[X_A\right] = \left\langle \psi\left[X_{[d]}\right], \psi^*\left[X_{[d]}\right]\right\rangle_{[X_A]} .$$

Here $\psi^*\left[X_{[d]}\right]$ is the complex conjugate of $\psi\left[X_{[d]}\right]$. When ψ is prepared by a quantum circuit acting on a initial state, the complex conjugate is the hermitean conjugate of the circuit acting on the complex conjgate of the initial state.

We abbreviate these contractions by extending the contraction diagrams with measurement symbols (see Figure 2.1).

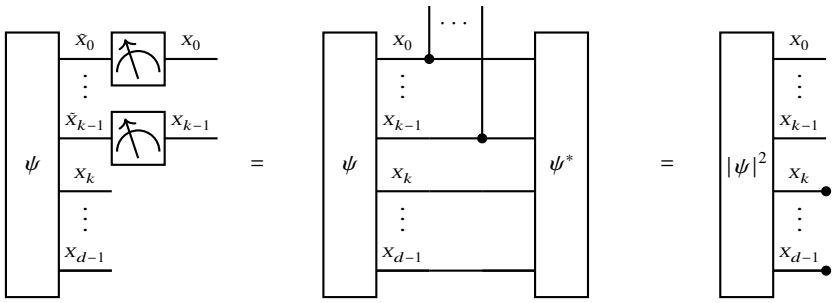


Fig. 2.1 Computational Basis Measurement of a quantum state ψ. The measurement symbols on the left side indicate the measured qubits and the first equation is understood as a definition. In the second equation we sketch, that the measurement distribution is equal to the contraction of the square absolute transform of ψ to the measured variables.

Each complex-valued tensor $\psi\left[X_{[d]}\right]$ has a decomposition into a phase tensor $\phi\left[X_{[d]}\right]$ and an absolute tensor $|\psi|\left[X_{[d]}\right]$ defined by

$$\psi\left[X_{[d]}\right] = \left\langle \exp\left[i \cdot \phi\left[X_{[d]}\right]\right], |\psi\left[X_{[d]}\right]|\right\rangle_{\left[X_{[d]}\right]} .$$

The measurement distribution depends only on $|\phi|$, that is

$$\mathbb{P}\left[X_{[d]}\right] = \left|\psi\left[X_{[d]}\right]\right|^2 .$$

Note, that this measurement distribution is not sensitive to the phases of the amplitudes.

When only a subset of variables is measured, the distribution is the contraction of the absolute square transform (these operations do not commute)

$$\mathbb{P}\left[X_{\mathcal{U}}\right] = \left\langle \left|\psi\left[X_{[d]}\right]\right|^2 \right\rangle_{[X_{\mathcal{U}}]} .$$

When we are interested in the preparation of quantum states with a specific computational basis measurement distribution, we can restrict to states with vanishing phase cores, that is

$$\psi\left[X_{[d]}\right] = \left\langle \exp\left[i \cdot 0\left[X_{[d]}\right]\right], |\psi|\left[X_{[d]}\right]\right\rangle_{\left[X_{[d]}\right]} = |\psi|\left[X_{[d]}\right] .$$

2.1.4.2 Phase sensitive measurement

When applying a Walsh-Hadamard transform on the qubits $\mathcal{V}\backslash\mathcal{U}$ to be closed (see Figure 2.2), the probability of measuring the closed qubits in the ground state is

$$\mathbb{P}\left[X_{\mathcal{V}\setminus\mathcal{U}} = 0_{\mathcal{V}\setminus\mathcal{U}}, X_{\mathcal{U}}\right] = \frac{1}{2^{|\mathcal{U}|}} \cdot \left|\left\langle \psi\left[X_{[d]}\right]\right\rangle_{[X_{\mathcal{U}}]}\right|^2 .$$

We thus sample from the contracted quantum state. Note that in this way, the relative phases of the amplitudes are contracted before the absolutes are taken. In this way, the measurement distribution is sensitive to the relative phases. This property is exploited in the Deutsch-Jozsa algorithm, which utilizes signs of the amplitudes.

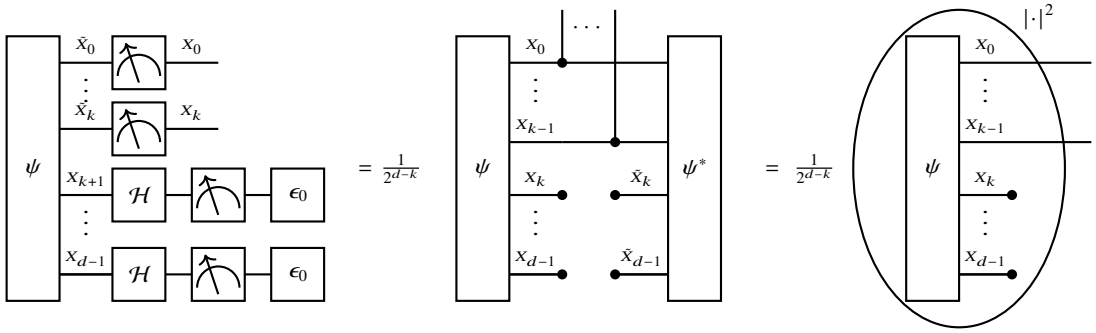


Fig. 2.2 Computational Basis Measurement of a quantum state ψ, after the Walsh-Hadamard Transform on the closed qubits. By the ellipse we indicate a coordinatewise transform to the square of the absolute, after the contraction.

2.1.4.3 POVM measurements

The most general measurement formalism is through a POVM, a set

$$\{E_y[X^{\text{in}}, X^{\text{out}}] \,:\, y \in [r]\}$$

of positive operators with

$$\sum_{y\in[r]} E_y[X^{\text{in}}, X^{\text{out}}] = \delta\left[X^{\text{in}}, X^{\text{out}}\right]$$

Measuring a pure state $|\psi\rangle$ we get the outcome y with probability

$$\left\langle \psi\left[X^{\text{out}}\right], E_y[X^{\text{in}}, X^{\text{out}}], \psi\left[X^{\text{in}}\right]\right\rangle_{[\varnothing]}$$

Introducing Y as an enumeration we define the tensor

$$E[Y, X^{\text{in}}, X^{\text{out}}] = \sum_{y\in[r]} \epsilon_y[Y] \otimes E_y[X^{\text{in}}, X^{\text{out}}] .$$

The condition on E is then

$$\left\langle E[Y, X^{\text{in}}, X^{\text{out}}]\right\rangle_{\left[X^{\text{in}}, X^{\text{out}}\right]} = \delta\left[X^{\text{in}}, X^{\text{out}}\right] .$$

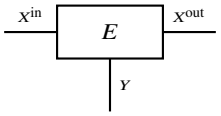
The measurement distribution of a pure state is then

$$\left\langle \psi\left[X^{\text{out}}\right], E[Y, X^{\text{in}}, X^{\text{out}}], \psi\left[X^{\text{in}}\right]\right\rangle_{[Y]} .$$

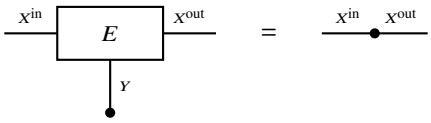
When having a mixed state defined by a density matrix $\rho[X^{\text{in}}, X^{\text{out}}]$, the distribution is

$$\left\langle E[Y, X^{\text{in}}, X^{\text{out}}], \rho[X^{\text{in}}, X^{\text{out}}]\right\rangle_{[Y]} .$$

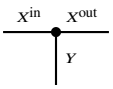
In our diagramatic picture, POVMs are tensors



such that



The standard computational basis measurement is



and satisfies these criteria.

2.2 Directed Circuits

We introduce a class of quantum circuits, which we frame directed circuits. We show that their measurement distribution is equivalent with that of Bayesian networks.

2.2.1 Definition

We build on the definition of uniformly controlled unitaries (see [21]), and study their alignment along directed acyclic hypergraphs. We define directed circuits using hypergraphs, which nodes are associated with the qubits of the circuit. The hyperedges are tuples of two subsets of the qubits, namely the incoming qubits representing control and the outgoing qubit representing the target of a unitary. To each hyperedge we then associate a uniformly controlled unitary.

Definition 2.1 (Directed Circuit) Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a directed acyclic hypergraph, where each hyperedge has exactly one outgoing node and all nodes appear exactly once as outgoing nodes of an hyperedge. Then a directed circuit on $\mathcal{G}$ is a decoration of the edges $e = (e^{\text{in}}, \{v\}) \in \mathcal{E}$ by uniformly controlled unitaries

$$\mathcal{U}^e \left[X^v_{\text{in}}, X^v_{\text{out}}, X^{e^{\text{in}}}_{\text{out}} \right] .$$

Since the hypergraph $\mathcal{G}$ to a directed circuit is acyclic, we find an order of the qubits, such that to each qubit appearing in the outgoing qubits of a hyperedge is ordered after the incoming qubits. In such way, we have a unique definition of the directed circuit as a concatenation of the controlled unitaries respecting a directionality order. When there are multiple orders respecting the directionality, then the concatenation respecting the orders all lead to an equivalent circuit.

Example 2.1 (Generic State Representation) In [21] a preparation scheme for arbitrary states with real and positive amplitudes by directed circuits is presented. Therein, the hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed as follows:

- Nodes $\mathcal{V} = [d]$
- Edges $\mathcal{E} = \{([k], \{k\}) \,:\, k \in [d]\}$

It is possible to realize each controlled unitary by a sequence of single qubit rotations and C-NOTs (see [21]), where the angles are computed using gray codes.

2.2.2 Preparation of q-samples to Bayesian Networks

2.2.2.1 Activation circuits

We now study circuits preparing q-samples of Bayesian Networks. To this end we introduce activation circuits, which are uniformly controlled unitaries (see [21]), where the rotation axis is chosen as the y-axes of the Bloch sphere, and the angles are computed by the function $h(\cdot)$ acting on a function value to be encoded.

We define the angle preparing function on $p \in [0,1]$ by

$$h(p) = 2 \cdot \cos^{-1}\left(\sqrt{1-p}\right) .$$

For any $p \in [0,1]$ we then have

$$\left\langle \epsilon_0\left[A^{\text{in}}\right], \mathcal{R}_Y\left(h(p)\right)\left[A^{\text{in}}, A^{\text{out}}\right]\right\rangle_{[A^{\text{out}}]} = \begin{pmatrix}\sqrt{1-p}\\ \sqrt{p}\end{pmatrix}\left[A^{\text{out}}\right] .$$

Definition 2.2 (Activation circuit) Given a function

$$f : \bigtimes_{k\in[d]} [m_k] \to [0,1]$$

its activation circuit is the uniformly controlled unitary $\mathcal{V}^f\left[A^{\text{in}}, A^{\text{out}}, X_{[d]}\right]$ defined as

$$\mathcal{V}^\tau\left[A^{\text{in}}, A^{\text{out}}, X_{[d]}\right] = \sum_{x_{[d]}\in\bigtimes_{k\in[d]}[m_k]} \epsilon_{x_{[d]}}\left[X_{[d]}\right] \otimes \mathcal{R}_Y\left(h(f\left[x_{[d]}\right])\right)\left[A^{\text{in}}, A^{\text{out}}\right] .$$

When we have a probability tensor, its activation circuit can be prepared, since all values are in $[0,1]$. Note that for rejection sampling, only the quotients of the values are important, we can therefore scale the value by a scalar such that the mode is 1.

Tensors $\tau\left[X_{[d]}\right]$ with non-negative coordinates can be encoded after dividing them by their maximum, that is the activation circuit of the function

$$f\left[x_{[d]}\right] = \frac{\tau\left[X_{[d]} = x_{[d]}\right]}{\max_{\tilde{y}_{[p]}} \tau\left[X_{[d]} = \tilde{y}_{[p]}\right]}$$

When the maximum of the tensor is not known, it can be replaced by an upper bound (reducing the acceptance rate of the rejection sampling).

2.2.2.2 Q-samples of Bayesian Networks

Q-samples of Bayesian Networks can be prepared by quantum circuits, where each conditional probability distribution is prepared by an activation circuit.

Theorem 2.1 ([19]) *Let there be a Bayesian Network $\mathbb{P}\left[X_{[d]}\right]$ on a directed acyclic hypergraph $\mathcal{G} = ([d], \mathcal{E})$ with dimensions $m_k = 2$ and decorations of the edges $\mathcal{E} = \{(\text{Pa}(k), \{k\}) : k \in [d]\}$ by conditional probabilities*

$$\mathbb{P}\left[X_k \middle| X_{\text{Pa}(k)}\right] .$$

To each $k \in [d]$ we construct the function

$$q_k : \bigtimes_{\tilde{k}\in\text{Pa}(k)} [2] \to [2] \quad , \quad q_k(x_{\text{Pa}(k)}) = \mathbb{P}\left[X_k = 1 \middle| X_{\text{Pa}(k)} = x_{\text{Pa}(k)}\right] .$$

Then the concatenation of the activation circuits (see Figure 2.3)

$$\mathcal{V}^{q_k}\left[X_k^{\text{in}}, X_k, X_{\text{Pa}(k)}\right]$$

respecting the order $[d]$ *prepares the q-sample to the Bayesian Network* $\mathbb{P}\left[X_{[d]}\right]$ *when acting on the initial state* $\bigotimes_{k\in[d]} \epsilon_0\left[X_k^{\text{in}}\right]$.

Proof. Let $\psi\left[X_{[d]}\right]$ be the state prepared by the activation circuits to the conditional probability distributions. We need to show that the measurement distribution of ψ is the Bayesian Network. For any $x_{[d]} \in \bigtimes_{k\in[d]}[2]$ we have

$$\begin{aligned}\left|\psi\left[X_{[d]} = x_{[d]}\right]\right|^2 &= \prod_{k\in[d]}\left(\mathcal{V}^{q_k}\left[X_k^{\text{in}} = 0, X_k = x_k, X_{\text{Pa}(k)} = x_{\text{Pa}(k)}\right]\right)^2 \\ &= \prod_{k\in[d]} \mathbb{P}\left[X_k = x_k \middle| X_{\text{Pa}(k)} = x_{\text{Pa}(k)}\right] \\ &= \mathbb{P}\left[X_{[d]} = x_{[d]}\right] .\end{aligned}$$

□

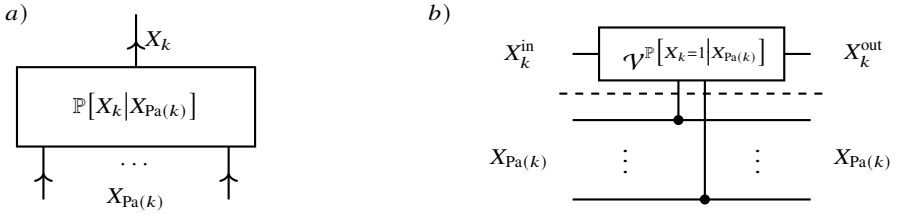


Fig. 2.3 Representation of directed and positive tensor by a controlled rotation. a) Conditional probability tensor $\mathbb{P}\left[X_k \middle| X_{\text{Pa}(k)}\right]$ being a tensor in a Bayesian Network. b) Circuit Encoding as a controlled rotation, which is the Activation circuit of the tensor $\mathbb{P}\left[X_k = 1 \middle| X_{\text{Pa}(k)}\right]$.

Example 2.2 (Encoding of discretized distributions) We here review the preparation of a discretized distribution from [13] as an application of activation circuits. Let $\mathbb{Q}[X]$ be a continuous probability distribution of a variable X taking values in $[0, 1]$. For any $d \in \mathbb{N}$ we define its discretization of order d as a discrete probability distribution $\mathbb{P}\left[X_{[d]}\right]$ with coordinates

$$\mathbb{P}^d\left[X_{[d]} = x_{[d]}\right] = \mathbb{Q}\left[X \in [\sum_{k\in[d]} 2^{-(k+1)}x_k, 2^{-d} + \sum_{k\in[d]} 2^{-(k+1)}x_k]\right] .$$

Then we have for any $p \in \mathbb{N}$ with $p \leq d$

$$\mathbb{P}^p\left[X_{[p]}\right] = \left\langle \mathbb{P}\left[X_{[d]}\right]\right\rangle_{\left[X_{[p]}\right]} .$$

The discretization order is thus increased by the contraction

$$\mathbb{P}^{d+1}\left[X_{[d+1]}\right] = \left\langle \mathbb{P}^d\left[X_{[d]}\right], \mathbb{P}^{d+1}\left[X_d | X_{[d]}\right]\right\rangle_{\left[X_{[d+1]}\right]} .$$

Having an activation circuit preparing $\mathbb{P}^d$ we can therefore prepare $\mathbb{P}^{d+1}$ by adding the activation circuit for the function

$$x_{[d]} \to \mathbb{P}^{d+1}\left[X_d = 1 | X_{[d]} = x_{[d]}\right] .$$

Note that the function evaluation amounts to the integration of $\mathbb{Q}$ as

$$\mathbb{P}^{d+1}\left[X_d = 1 | X_{[d]} = x_{[d]}\right] = \frac{\mathbb{Q}\left[X \in [2^{-(d+1)} + \sum_{k\in[d]} 2^{-(k+1)}x_k, 2^{-d} + \sum_{k\in[d]} 2^{-(k+1)}x_k]\right]}{\mathbb{Q}\left[X \in [\sum_{k\in[d]} 2^{-(k+1)}x_k, 2^{-d} + \sum_{k\in[d]} 2^{-(k+1)}x_k]\right]} .$$

2.2.3 Equivalence with Bayesian Networks

Above we have shown, how Bayesian Networks can be prepared by activation circuits, which are a class of directed circuits. We now show, that the measurement distribution of any directed circuit is a Bayesian Network. It then follows that the set of distributions prepared by directed circuits is exactly the set of Bayesian Networks.

This is closely connected to the chain decomposition of probability distributions, and their construction. In [19] Bayesian Networks are prepared by directed circuits. When in the graph $\mathcal{G}$ each node appears at most once as outgoing node, it is also the hypergraph to a family of Bayesian Networks. The measurement distributions of a state prepared by a $\mathcal{G}$-controlled Circuit acting on disentangled initial states are exactly the Bayesian Networks with respect to $\mathcal{G}$.

Theorem 2.2 *Let $\mathcal{G}$ be a directed acyclic hypergraph, such that node appears at most once as an outgoing node. The measurement distributions of the by $\mathcal{G}$ controlled circuits acting on disentangled initial states are equal to the Bayesian Networks on $\mathcal{G}$.*

Lemma 2.1 *Let $\mathcal{G}$ be a directed acyclic hypergraph, such that node appears at most once as an outgoing node. Then any Bayesian network on $\mathcal{G}$ can be prepared by a $\mathcal{G}$-controlled circuit with activation circuits of the conditional probability tensors.*

Proof. Let $\mathbb{P}\left[X_{[d]}\right]$ be a Bayesian network on the graph $\mathcal{G}$. Enumerate the nodes $\mathcal{V}$ of the $\mathcal{G}$ by $[d]$, such that for each $k \in [d]$ we have $\mathrm{Pa}(k) \subset [k]$. Then define a $\mathcal{G}$-controlled circuit, by choosing for each $k \in [d]$ controlled unitaries which satisfy

$$\mathcal{U}^k \left[X_k^{\mathrm{in}} = 0, X_{v,\mathrm{out}}, X_{\mathrm{Pa}(k),\mathrm{out}}\right] = \sqrt{\mathbb{P}\left[X_k \middle| X_{\mathrm{Pa}(k)}\right]}\,.$$

Here we specified only the action of the controlled unitary on the basis vector $\epsilon_0\left[X_k\right]$, the action on $\epsilon_1\left[X_k\right]$ can be chosen by an arbitrary orthogonal unit vector. For more explicit construction, see the activation circuits. Any such defined $\mathcal{G}$-controlled circuit acting on the initial state $\bigotimes_{k\in[d]} \epsilon_0\left[X_k\right]$ prepares a quantum state $\psi\left[X_{[d]}\right]$ with measurement distribution

$$|\psi|^2\left[X_{[d]}\right]\,.$$

Given arbitrary $x_{[d]} \in \bigtimes_{k\in[d]}[m_k]$ we have

$$|\psi|^2\left[X_{[d]} = x_{[d]}\right] = \prod_{k\in[d]} \mathbb{P}\left[X_k = x_k \middle| X_{\mathrm{Pa}(k)} = x_{\mathrm{Pa}(k)}\right] = \mathbb{P}\left[X_{[d]} = x_{[d]}\right]\,.$$

Here we used in the last equation, that $\mathbb{P}\left[X_{[d]}\right]$ is a Bayesian network. Since the equivalence holds for any coordinate, this establishes the equivalence of the measurement distribution of $\psi\left[X_{[d]}\right]$ and $\mathbb{P}\left[X_{[d]}\right]$. □

While we have already shown by Lem. 2.1 that arbitrary Bayesian Networks can be prepared by directed circuits, we now show the converse, namely that any distribution prepared by directed circuits is a Bayesian Network. To this end, we use the characterization of Bayesian Networks by the conditional independencies they encode (see [18]). To be more precise, we need to show that any node variable is conditionally independent of its non-descendants given its parents.

Lemma 2.2 *Let $(\mathcal{G},\mathcal{U})$ be a $\mathcal{G}$-controlled circuit acting on a disentangled initial state and $\mathbb{P}\left[X_{\mathcal{V}}\right]$ the corresponding measurement distribution. Then we have for each $v \in \mathcal{V}$ the conditional independence*

$$\left(X_v \perp X_{\mathrm{NonDes}(v)}\right) \,\middle|\, X_{\mathrm{Pa}(v)}\,.$$

Proof. We choose to a given $v \in \mathcal{V}$ an enumeration $[d]$ of the nodes, such that for each $k \in [d]$ we have $\mathrm{Pa}(k) \subset [k]$ and for the enumerator $\tilde{k}$ of v we further have $\mathrm{NonDes}(\tilde{k}) \subset [\tilde{k}]$. Let $\mathbb{P}\left[X_{[d]}\right]$ be the measurement distribution of the $\mathcal{G}$-controlled circuit acting on a disentangled initial state $\bigotimes_{k\in[d]} \psi^k\left[X_k\right]$ and choose arbitrary $x_{[k]}$. We then have

$$\begin{aligned}\mathbb{P}\left[X_{\tilde{k}}, X_{[k]} = x_{[k]}\right] &= \left\langle \left(\bigcup_{k\in[d]} \{\mathcal{U}^k, \mathcal{U}^{k,\dagger}, \psi^k, \psi^{k,*}\} \right) \cup \left(\bigcup_{k\in[\tilde{k}]} \epsilon_{x_k}\left[X_k^{\text{out}}\right] \right) \right\rangle_{[X_{\tilde{k}}]} \\ &= \left\langle \bigcup_{k\in[\tilde{k}]} \{\mathcal{U}^k, \mathcal{U}^{k,\dagger}, \psi^k, \psi^{k,*}, \epsilon_{x_k}\left[X_k^{\text{out}}\right]\} \right\rangle_{[X_{\tilde{k}}]} \\ &= \left| \left\langle \mathcal{U}^{\tilde{k}}\left[X_{\tilde{k},\text{in}}, X_{\tilde{k},\text{out}}\right], X_{\text{Pa}(\tilde{k}),\text{out}} = x_{\text{Pa}(\tilde{k}),\text{out}} \right\rangle_{[X_{\tilde{k},\text{out}}]} \right|^2 \\ &\quad \cdot \prod_{k\in[\tilde{k}]} \left(\left\langle \psi^k\left[X_k^{\text{in}}\right], \mathcal{U}^k\left[X_k^{\text{in}}, X_k^{\text{out}}, X_{\text{Pa}(k),\text{out}} = x_{\text{Pa}(k),\text{out}}\right] \right\rangle_{[\varnothing]} \right)^2\end{aligned}$$

Here we used in the second equation the unitarity of the controlled unitaries to $k \notin [\tilde{k}]$. Since the indices $x_{[\tilde{k}]/\text{Pa}(\tilde{k})} = x_{\text{NonDes}(\tilde{k})}$ appear only in the constant term, we conclude

$$\mathbb{P}\left[X_{\tilde{k}} \middle| X_{[k]}\right] = \mathbb{P}\left[X_{\tilde{k}} \middle| X_{\text{Pa}(\tilde{k})}\right] \otimes \mathbb{I}\left[X_{\text{NonDes}(\tilde{k})}\right] ,$$

which establishes the conditional independence $\left(X_v \perp X_{\text{NonDes}(v)}\right) \middle| X_{\text{Pa}(v)}$. □

Proof of Thm. 2.2. The theorem follows directly from Lem. 2.1 and Lem. 2.2, using that Bayesian Networks are characterized by the conditional independence of each variable to its non-descendants given its parents. □

Example 2.3 (Chain decomposition) The hypergraph in Example 2.1 corresponds with a family of Bayesian Networks subsuming any probability distribution. This can be verified based on the chain decomposition of a generic distribution.

Another question is, whether each quantum state, which measurement distribution is a Bayesian Network can be prepared by a $\mathcal{G}$-controlled circuit. This is not always the case, since the phase tensor does not influence the measurement distribution. Any phase tensor, of a by $\mathcal{G}$-controlled circuit prepared state has however a decomposition

$$\phi\left[X_{[d]}\right] = \sum_{k\in[d]} \phi^k\left[X_k, X_{\text{Pa}(k)}\right] \otimes \mathbb{I}\left[X_{[d]/\{\{k\}\cup\text{Pa}(k)\}}\right] ,$$

where the phase cores ϕ^k can be read of the controlled unitaries. When there are phase cores which do not have such a decomposition, the corresponding states are not representable. [21] shows, that when the graph is chosen as in Example 2.1, then also the phases can be represented.

2.2.4 Ancilla Augmentation

We provide here some further theory underlying the ancilla augmentation of Computation-Activation Networks.

Graph controlled circuits prepare Bayesian Networks, and do not apply to undirected graphical models such as Markov Networks and Computation-Activation Networks. We therefore introduce in the following ancilla augmentation, which is a method to represent Markov Networks as conditioned Bayesian Network. This enables usage of the previously used preparation schemes for Bayesian Networks.

2.2.4.1 Post-selection by Ancilla Variables

First of all we define ancilla augmentation and then show how they can be prepared as measurement distributions using activation circuits.

Definition 2.3 (Ancilla Augmentation of a Distribution) Let $\mathbb{P}\left[X_{[d]}\right]$ be a probability distribution of variables $X_{[d]}$. Another distribution $\tilde{\mathbb{P}}[X_{[d]}, A_{[p]}]$ of variables $X_{[d]}$ and ancilla variables $A_{[p]}$ is called an ancilla augmentation of $\mathbb{P}\left[X_{[d]}\right]$, if there is $\lambda > 0$ with

$$\lambda \cdot \mathbb{P}\left[X_{[d]}\right] = \tilde{\mathbb{P}}[X_{[d]}, A_{[p]} = 1_{[p]}] \,.$$

We refer to λ as the corresponding acceptance probability.

Note that $\lambda \leq 1$ since

$$\lambda = \left\langle \lambda \cdot \mathbb{P}\left[X_{[d]}\right] \right\rangle_{[\varnothing]} = \left\langle \tilde{\mathbb{P}}[X_{[d]}, A_{[p]} = 1_{[p]}] \right\rangle_{[\varnothing]} \leq \left\langle \tilde{\mathbb{P}}[X_{[d]}, A_{[p]}] \right\rangle_{[\varnothing]} = 1 \,.$$

Sampling from the distribution can be done by rejection sampling, also called post selection, on the ancilla augmented distribution in the following way. When drawing a sample $(x_{[d]}, a)$ from $\tilde{\mathbb{P}}[X_{[d]}, A]$ and rejecting it whenever $a = 0$, the accepted samples are distributed as $\mathbb{P}\left[X_{[d]}\right]$. We then have λ as the probability of accepting a sampling in this rejection sampling method. The acceptance probability is critical in determining the efficiency of this approach, since we need to draw $\sim \frac{1}{\lambda}$ samples from $\tilde{\mathbb{P}}[X_{[d]}, A]$ to get an accepted sample.

Since

$$\tilde{\mathbb{P}}[X_{[d]}, A] = \left\langle \tilde{\mathbb{P}}[X_{[d]}], \tilde{\mathbb{P}}[A|X_{[d]}] \right\rangle_{\left[X_{[d]}, A\right]} \tag{2.1}$$

the described sampling procedure is equal to drawing $x_{[d]}$ from the marginal

$$\tilde{\mathbb{P}}[X_{[d]}]$$

and accepting the sample with probability

$$\tilde{\mathbb{P}}[A = 1|X_{[d]}] \,.$$

2.2.4.2 Construction based on Uniform Distributions

We construct ancilla augmentations based on the decomposition (2.1) where the marginal of $X_{[d]}$ is the uniform distribution, that is

$$\tilde{\mathbb{P}}[X_{[d]}] = \frac{1}{\prod_{k\in[d]} m_k} \mathbb{I}\left[X_{[d]}\right]$$

and we construct the conditional probability of the ancilla variable such that

$$\tilde{\mathbb{P}}[A = 1|X_{[d]}] = \lambda \cdot \left(\prod_{k\in[d]} m_k\right) \mathbb{P}\left[X_{[d]}\right] \,.$$

For the latter we can apply the preparation schemes of conditioned distributions, which have been studied above for Bayesian Networks.

We denote for any non-negative tensor $\tau\left[X_{[d]}\right]$ we construct

$$\begin{aligned}\mathcal{A}^{\tau,\lambda}\left[A, X_{[d]}\right] = \sum_{x_{[d]} \in \bigtimes_{k\in[d]}[m_k]} \epsilon_{x_{[d]}}\left[X_{[d]}\right] \otimes \Big(&\lambda \cdot \big(\prod_{k\in[d]} m_k\big) \cdot \tau\left[X_{[d]} = x_{[d]}\right] \cdot \epsilon_1\left[A\right] \\ &+ \big(1 - \lambda \cdot \big(\prod_{k\in[d]} m_k\big) \cdot \tau\left[X_{[d]} = x_{[d]}\right]\big) \cdot \epsilon_0\left[A\right]\Big) \,.\end{aligned}$$

For this to be a conditional distribution we need to demand that

$$\max_{x_{[d]} \in \bigtimes_{k\in[d]}[m_k]} \lambda \cdot \big(\prod_{k\in[d]} m_k\big) \cdot \tau\left[X_{[d]} = x_{[d]}\right] \leq 1 \,,$$

which is equivalent to

$$\lambda \leq \left(\prod_{k\in[d]} m_k\right)^{-1} \cdot \left(\max_{x_{[d]}\in\bigtimes_{k\in[d]}[m_k]} \tau\left[X_{[d]} = x_{[d]}\right]\right)^{-1}.$$

Note that if $\tau\left[X_{[d]}\right]$ is the uniform distribution $\mathbb{I}\left[X_{[d]}|\varnothing\right]$, then we can achieve the best acceptance rate $\lambda = 1$.

Given a distribution $\mathbb{P}\left[X_{[d]}\right]$ we add an ancilla variable A and construct the augmented distribution (see Figure 2.4)

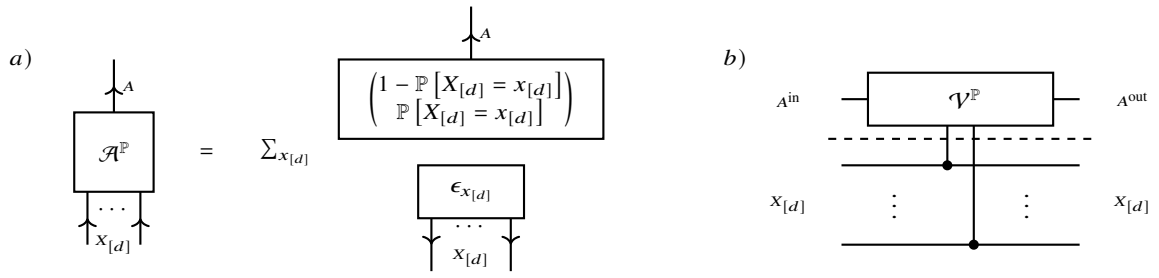


Fig. 2.4 Ancilla augmentation of a distribution $\mathbb{P}\left[X_{[d]}\right]$. a) Augmented distribution $\mathcal{A}^{\mathbb{P}}\left[A, X_{[d]}\right]$ with the property that $\mathbb{P}\left[X_{[d]}\right] = \mathcal{A}^{\mathbb{P}}\left[X_{[d]}|A = 1\right]$. b) Preparation of the augmented distribution by the activation circuit of $\mathbb{P}\left[X_{[d]}\right]$.

Lemma 2.3 *The activation circuit to a probability distribution $\mathbb{P}\left[X_{[d]}\right]$ applied on the initial state*

$$\sqrt{\frac{1}{\prod_{k\in[d]} m_k}}\mathbb{I}\left[X_{[d]}\right] \otimes \epsilon_0\left[A\right]$$

prepares a quantum state ψ, which is the Q-sample of the ancilla augmentation

$$\left\langle \mathcal{A}^{\mathbb{P},\frac{1}{m}}\left[A, X_{[d]}\right]\right\rangle_{\left[A,X_{[d]}|\varnothing\right]}$$

with the acceptance probability $m^{-1} := \left(\prod_{k\in[d]} m_k\right)^{-1}$.

Proof. We first show, that $\left\langle \mathcal{A}^{\mathbb{P},\frac{1}{m}}\left[A, X_{[d]}\right]\right\rangle_{\left[A,X_{[d]}|\varnothing\right]}$ is indeed an ancilla augmentation. We have

$$\lambda = \left\langle \mathcal{A}^{\mathbb{P}}\left[A, X_{[d]}\right]\right\rangle_{[A=1]} = \frac{1}{m}\sum_{x_{[d]}\in\bigtimes_{k\in[d]}[m_k]} \mathbb{P}\left[X_{[d]} = x_{[d]}\right] = \frac{1}{m}.$$

and for any $x_{[d]} \in \bigtimes_{k\in[d]}[m_k]$ it holds that

$$\left\langle \mathcal{A}^{\mathbb{P},\frac{1}{m}}\left[A, X_{[d]}\right]\right\rangle_{\left[A=1,X_{[d]}=x_{[d]}|\varnothing\right]} = \frac{1}{\lambda}\frac{1}{\prod_{k\in[d]} m_k}\mathbb{P}\left[X_{[d]} = x_{[d]}\right] = \mathbb{P}\left[X_{[d]} = x_{[d]}\right],$$

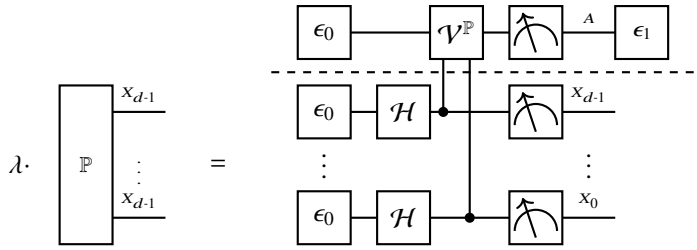
The tensor $\left\langle \mathcal{A}^{\mathbb{P},\frac{1}{m}}\left[A, X_{[d]}\right]\right\rangle_{\left[A,X_{[d]}|\varnothing\right]}$ is thus an ancilla augmentation for $\mathbb{P}$.

We continue to show that the quantum state ψ prepared by the activation circuit is the Q-sample of $\left\langle \mathcal{A}^{\mathbb{P},\frac{1}{m}}\left[A, X_{[d]}\right]\right\rangle_{\left[A,X_{[d]}|\varnothing\right]}$. Then we have for any $x_{[d]} \in \bigtimes_{k\in[d]}[m_k]$

$$\psi\left[A, X_{[d]} = x_{[d]}\right] = \sqrt{\frac{1}{\prod_{k\in[d]} m_k}}\left(\sqrt{\mathbb{P}\left[X_{[d]} = x_{[d]}\right]}\cdot\epsilon_1\left[A\right] + \sqrt{\left(1 - \mathbb{P}\left[X_{[d]} = x_{[d]}\right]\right)}\cdot\epsilon_0\left[A\right]\right),$$

which is indeed the Q-sample. □

In our tensor network circuit notation, Lem. 2.3 states the equation



for the acceptance probability (see proof of Lem. 2.3)

$$\lambda = \frac{1}{\prod_{k\in[d]} m_k} .$$

Note that the acceptance probability is exponentially small in the number of variables. If $\mathbb{P}\left[X_{[d]}\right]$ is not uniform, we can improve by building the activation circuit to the function

$$\frac{1}{\max_{x_{[d]}\in\bigtimes_{k\in[d]}[m_k]} \mathbb{P}\left[X_{[d]} = x_{[d]}\right]} \cdot \mathbb{P}\left[X_{[d]}\right] .$$

2.2.4.3 Best Acceptance Probability by ∞-Renyi Divergence

Let us now show, how the best acceptance probability in sets of ancilla augmented distributions is connected with ∞-Renyi divergences. For $0 < \alpha < \infty$ and $\alpha \neq 1$ the Renyi Divergence is defined as

$$D_\alpha\left[\mathbb{P}||\mathbb{Q}\right] = \frac{1}{1-\alpha} \ln\left[\sum_{x_{[d]}\in\bigtimes_{k\in[d]}[m_k]} \frac{\mathbb{P}\left[X_{[d]} = x_{[d]}\right]^\alpha}{\mathbb{Q}\left[X_{[d]} = x_{[d]}\right]^{\alpha-1}}\right] .$$

The ∞ Renyi Divergence is the limit $\alpha \to \infty$

$$D_\infty\left[\mathbb{P}||\mathbb{Q}\right] = \ln\left[\max_{x_{[d]}\in\bigtimes_{k\in[d]}[m_k]} \frac{\mathbb{P}\left[X_{[d]} = x_{[d]}\right]}{\mathbb{Q}\left[X_{[d]} = x_{[d]}\right]}\right] .$$

Theorem 2.3 *Let $\mathbb{P}\left[X_{[d]}\right]$ and $\mathbb{Q}[X_{[d]}]$ be distributions, such that for each $x_{[d]} \in \bigtimes_{k\in[d]}[m_k]$ with $\mathbb{P}\left[X_{[d]} = x_{[d]}\right] > 0$ we also have $\mathbb{Q}[X_{[d]} = x_{[d]}] > 0$. Let us consider all ancilla augmentations $\tilde{\mathbb{P}}[X_{[d]}, A]$ which marginal is $\mathbb{Q}[X_{[d]}]$, that is*

$$\mathbb{Q}[X_{[d]}] = \left\langle \tilde{\mathbb{P}}[X_{[d]}, A]\right\rangle_{[X_{[d]}]} .$$

The best acceptance probability among these ancilla augmentations is then

$$\lambda = \exp\left[-D_\infty\left[\mathbb{P}||\mathbb{Q}\right]\right] ,$$

and achieved when for any $x_{[d]} \in \bigtimes_{k\in[d]}[m_k]$

$$\tilde{\mathbb{P}}[A = 1|X_{[d]} = x_{[d]}] = \frac{\mathbb{P}\left[X_{[d]} = x_{[d]}\right]}{\mathbb{Q}\left[X_{[d]} = x_{[d]}\right]} \cdot \left(\max_{x_{[d]}\in\bigtimes_{k\in[d]}[m_k]} \frac{\mathbb{P}\left[X_{[d]} = x_{[d]}\right]}{\mathbb{Q}\left[X_{[d]} = x_{[d]}\right]}\right)^{-1} .$$

Proof. Any $\tilde{\mathbb{P}}[X_{[d]}, A]$ with marginal $\mathbb{Q}\left[X_{[d]}\right]$ has a decomposition

$$\tilde{\mathbb{P}}[X_{[d]}, A] = \left\langle \mathbb{Q}\left[X_{[d]}\right], \tilde{\mathbb{P}}[A|X_{[d]}]\right\rangle_{[X_{[d]}, A]}$$

and in order to be an ancilla augmentation for $\mathbb{P}\left[X_{[d]}\right]$ we find $\lambda > 0$ with

$$\tilde{\mathbb{P}}[X_{[d]}, A = 1] = \lambda \cdot \mathbb{P}\left[X_{[d]}\right] .$$

It follows that for each $x_{[d]} \in \bigtimes_{k\in[d]}[m_k]$

$$1 \geq \tilde{\mathbb{P}}[A = 1 | X_{[d]} = x_{[d]}] = \lambda \cdot \frac{\mathbb{P}\left[X_{[d]} = x_{[d]}\right]}{\mathbb{Q}\left[X_{[d]} = x_{[d]}\right]} .$$

and thus

$$\lambda \leq \left(\max_{x_{[d]}\in\bigtimes_{k\in[d]}[m_k]} \frac{\mathbb{P}\left[X_{[d]} = x_{[d]}\right]}{\mathbb{Q}\left[X_{[d]} = x_{[d]}\right]} \right)^{-1} . \tag{2.2}$$

For the in the claim constructed ancilla augmentation, we have the acceptance probability

$$\begin{aligned}
\tilde{\mathbb{P}}[A = 1] &= \mathbb{E}_{x_{[d]}\sim\mathbb{Q}} \left[\frac{\mathbb{P}\left[X_{[d]} = x_{[d]}\right]}{\mathbb{Q}\left[X_{[d]} = x_{[d]}\right]} \cdot \left(\max_{x_{[d]}\in\bigtimes_{k\in[d]}[m_k]} \frac{\mathbb{P}\left[X_{[d]} = x_{[d]}\right]}{\mathbb{Q}\left[X_{[d]} = x_{[d]}\right]} \right)^{-1} \right] \\
&= \left\langle \mathbb{Q}\left[X_{[d]}\right], \left(\mathbb{Q}\left[X_{[d]}\right]\right)^{-1}, \mathbb{P}\left[X_{[d]}\right] \right\rangle_{[\varnothing]} \left(\max_{x_{[d]}\in\bigtimes_{k\in[d]}[m_k]} \frac{\mathbb{P}\left[X_{[d]} = x_{[d]}\right]}{\mathbb{Q}\left[X_{[d]} = x_{[d]}\right]} \right)^{-1} \\
&= \left(\max_{x_{[d]}\in\bigtimes_{k\in[d]}[m_k]} \frac{\mathbb{P}\left[X_{[d]} = x_{[d]}\right]}{\mathbb{Q}\left[X_{[d]} = x_{[d]}\right]} \right)^{-1} \\
&= \exp\left[-D_\infty\left[\mathbb{P}||\mathbb{Q}\right]\right] .
\end{aligned}$$

The constructed ancilla augmented distribution satisfies the bound (2.2) on the acceptance probability holds straight, and it is thus the one with largest acceptance rate. □

Note, that in the section above we have constructed ancilla augmentations with uniform $\mathbb{Q}$ by activation circuits. For these the optimal acceptance rate is

$$\lambda = \left(\prod_{k\in[d]} m_k \right)^{-1} \cdot \left(\max_{x_{[d]}\in\bigtimes_{k\in[d]}[m_k]} \mathbb{P}\left[X_{[d]} = x_{[d]}\right] \right)^{-1} .$$

Thus, $\lambda = 1$ is only achievable if $\mathbb{P}\left[X_{[d]}\right]$ is itself the uniform distribution.

Extension: We could increase the acceptance probability, when we sample from a proposal distribution $\mathbb{Q}$ with smaller ∞ Renyi divergence to $\mathbb{P}$. When sampling with Quantum Circuits, this could be implemented by a state preparation for the distributed variables, before the Computation Activation Circuit is applied. It would be interesting to train variational quantum circuits for this task. However, when we want to apply the same scheme as above, one needs to encode $\mathbb{Q}$ into the ancilla preparing rotations, so $\mathbb{Q}$ would need to be an elementary Computation-Activation Network as well.

2.2.4.4 Augmentation of Markov Networks

Let us now show how multiple ancilla variables can be designed to sample from arbitrary Markov Networks.

Example 2.4 (Elementary Tensors) Let there be an elementary distribution

$$\mathbb{P}\left[X_{[d]}\right] = \bigotimes_{k\in[d]} \mathbb{P}^k\left[X_k\right] ,$$

that is, all variables are pairwise independent. Then the ancilla augmentation of the distribution is given by the Bayesian Network with the hypergraph shown in Figure 2.5a). Compared with the ancilla augmentation of a generic distribution this augmentation is sparser in the circuit depth (see Figure 2.5b), but introduces multiple ancilla variables. The sparsity originates since each ancilla variable is controlled by a single qubit. The total amount of controlled rotations is thus $\sum_{k\in[d]}[m_k]$ instead of $\prod_{k\in[d]}[m_k]$ for the generic implementation.

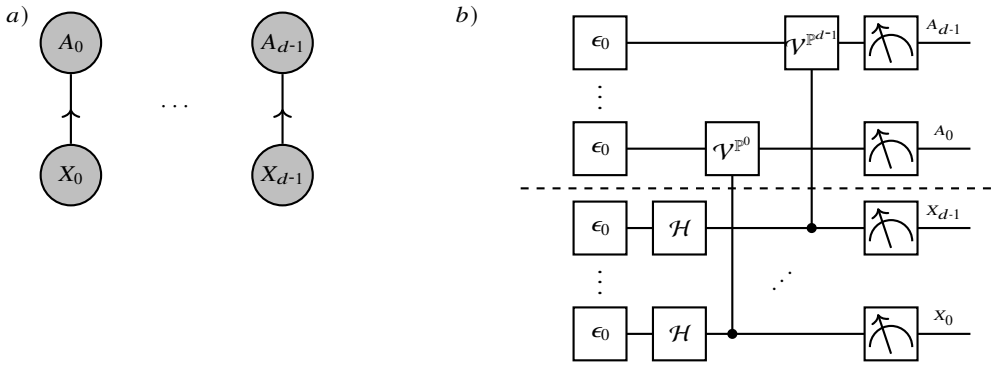


Fig. 2.5 Ancilla augmentation of an elementary distribution (which variables are pairwise independent). a) Hypergraph to the Bayesian Network representing the ancilla augmentation. b) Circuit preparing the ancilla augmented distribution by activation circuits acting on the disentangled initial state.

Let there be a Markov Network, i.e. any non-vanishing tensor network of non-negative tensors, which is understood as a probability disribution after normalization (see [11]). For an example of a Markov Network see Figure 2.6a.

We apply a similar ancilla augmentation on any hypercore of a Markov Network, as has been done above for probability distributions. To prepare, we need to scale each core by dividing by its maximum (or any upper bound on its maximum), such that any coordinate is smaller than 1. We introduce for each core an boolean ancilla variable A_e (while X_e is the tuple of distributed variables collected in the hyperedge e) then build directed tensors

$$\mathcal{A}^{\tau^e,\lambda^e}[A_e, X_e] = \sum_{x_e} \epsilon_{x_e}[X_e] \otimes \Big(\lambda^e \cdot \big(\prod_{k\in e} m_k\big) \cdot \tau^e[X_e = x_e] \cdot \epsilon_1[A] + \big(1 - \lambda^e \cdot \big(\prod_{k\in e} m_k\big) \cdot \tau^e[X_e = x_e]\big) \cdot \epsilon_0[A]\Big).$$

where we choose the maximal values

$$\lambda^e = \left(\prod_{k\in e} m_k\right)^{-1} \cdot \left(\max_{x_e} \tau^e[X_e = x_e]\right)^{-1},$$

such that $\mathcal{A}^e$ stays non-negative and can be interpreted as a conditional probability distribution. Note, that λ^e is not an acceptance rate itself, since τ^e is not always a distribution.

Lemma 2.4 *When augmenting all cores of a Markov Network with distinct ancilla variables in this way (see Figure 2.6b), we get an ancilla augmentation of the Markov Network.*

$$\tilde{\mathbb{P}}[A_\mathcal{E}, X_{[d]}] = \left\langle \{\mathcal{A}^{\tau^e,\lambda^e}[A_e, X_e] \,:\, e \in \mathcal{E}\} \cup \{\mathbb{I}\left[X_{[d]}|\varnothing\right]\}\right\rangle_{[A_\mathcal{E}, X_{[d]}]}.$$

Proof. $\tilde{\mathbb{P}}[A_\mathcal{E}, X_{[d]}]$ is a Bayesian Network (see Figure 2.6d) and thus a distribution. We further have

$$\begin{aligned}\tilde{\mathbb{P}}[A_\mathcal{E} = 1_\mathcal{E}, X_{[d]}] &= \left\langle \{\mathcal{A}^{\tau^e,\lambda^e}[A_e = 1, X_e] \,:\, e \in \mathcal{E}\} \cup \{\mathbb{I}\left[X_{[d]}|\varnothing\right]\}\right\rangle_{[A_\mathcal{E}, X_{[d]}]} \\ &= \frac{\prod_{e\in\mathcal{E}} \lambda^e \left(\prod_{k\in e} m_k\right)}{\prod_{k\in[d]} m_k} \langle \{\tau^e[X_e] \,:\, e \in \mathcal{E}\}\rangle_{[X_{[d]}]}.\end{aligned}$$

Thus, $\tilde{\mathbb{P}}[A_{\mathcal{E}}, X_{[d]}]$ is an ancilla augmentation of the Markov Network with acceptance probability

$$\frac{\prod_{e\in\mathcal{E}} \lambda^e \left(\prod_{k\in e} m_k\right)}{\langle\{\tau^e\left[X_e\right] : e\in\mathcal{E}\}\rangle_{[\varnothing]} \cdot \left(\prod_{k\in[d]} m_k\right)} = \left(\prod_{k\in[d]} m_k\right)^{-1} \cdot \frac{\prod_{e\in\mathcal{E}} \left(\max_{x_e} \tau^e\left[X_e = x_e\right]\right)}{\langle\{\tau^e\left[X_e\right] : e\in\mathcal{E}\}\rangle_{[\varnothing]}} .$$

□

The main advantage of this augmentation is, that all cores are directed (see Figure 2.6b), and build therefore a Bayesian Network. Using the activation circuits studied above, we can now determine a rejection sampling procedure for arbitrary Markov Networks.

Lemma 2.5 *Let there be a Markov Network $\tau^{\mathcal{G}} = \{\tau^e\left[X_e\right] : e\in\mathcal{E}\}$ on a hypergraph $\mathcal{G}$. To each $e\in\mathcal{E}$ we build a function*

$$q_e = \frac{\tau^e\left[X_e = y_e\right]}{\max_{y_e} \tau^e\left[X_e = y_e\right]} .$$

We then build a concatenation of activation circuits to each function q_e, where each has a distinct ancilla qubit A_e. Then the constructed directed circuit acting on the state

$$\sqrt{\frac{1}{\prod_{v\in\mathcal{V}} m_v}}\mathbb{I}\left[X_{\mathcal{V}}\right] \otimes \left(\bigotimes_{e\in\mathcal{E}} \epsilon_0\left[A_e\right]\right)$$

prepares an ancilla augmented distribution of the Markov Network.

Proof. We show, that the measurement distribution is the ancilla augmented Markov Network. □

We use this result to prepare samples from Markov Networks by rejection sampling on the ancilla variables, in qcreason (see Sect. **??**).

When the maximum of the hypercores are not known, and only an upper bound is given, it suffice to rescale them such that the maximum is lower than 1.

When the maximal coordinate of the Markov Network is, restricted to each edge, the maximal coordinate of the hypercore of that edge, we say that the Markov Network is not frustrated, otherwise frustrated.

Lemma 2.6 *If the Markov Network is not frustrated, then the acceptance probability λ of the rejection sampling scheme using the ancilla augmented Markov Network (with respect to the uniform distribution) is the ∞-Renyi divergence between the Markov Network and the uniform distribution.*

Proof. If and only if the Markov Network is not frustrated we have

$$\max_{x_{[d]}} \left(\prod_{e\in\mathcal{E}} \tau^e\left[X_e = x_e\right]\right) = \prod_{e\in\mathcal{E}} \left(\max_{x_e} \tau^e\left[X_e = x_e\right]\right) .$$

In this case, the ancilla augmentation has the best acceptance probability, since for the state $x_{[d]}$ maximizing all local cores we have

$$\tilde{\mathbb{P}}[A_{[p]} = 1_{[p]} | X_{[d]} = x_{[d]}] = 1 .$$

Alternatively: Show this with the acceptance rate in proof of Lem. 2.4, comparing with best naive augmentation. □

If the Markov Network is frustrated, we cannot achieve the best acceptance probability by the distributed ancilla preparation approach described above. In this case, one faces a tradeoff between the size of the circuit to prepare ancilla augmentations and the acceptance probability.

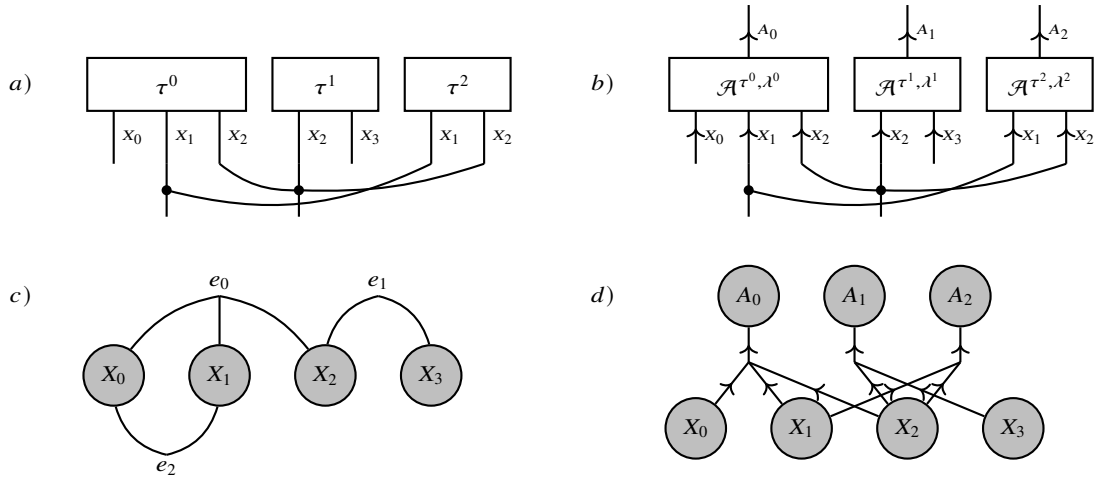


Fig. 2.6 Ancilla augmentation of a Markov Network a) onto a Bayesian Network b). To this end the arbitrary non-negative cores τ^e are replaced by directed ancilla augmented cores $\tilde{\tau}^e$, with ancilly variables outgoing.

2.3 Computation circuits

In the previous sections we have studied the schemes to encode arbitrary conditional probability distributions, which concatenation prepares q-samples of Bayesian networks [19]. While these schemes do not assume any further structure in the conditional probability distributions, we now investigate how to exploit the structure to achieve sparser circuits. To be more precise, we focus on deterministic conditional probability distributions, which are basis encodings to functions. To this end, let us first introduce a quantum pendant to encode functions, which has the decomposition by contraction property as we show later. We will also relate them to basis encodings.

Definition 2.4 (Computation circuit) Given a boolean function $f : \bigtimes_{k\in[d]}[2] \to [2]$ the computation circuit is the unitary tensor

$$\begin{aligned} C^f\left[Y_f^{\text{in}}, Y_f^{\text{out}}, X_{[d]}\right] \quad = & \sum_{x_{[d]}\in\bigtimes_{k\in[d]}[m_k]\,:\,f\left[x_{[d]}\right]=1} \sigma_1\left[Y_f^{\text{in}}, Y_f^{\text{out}}\right] \otimes \epsilon_{x_{[d]}}\left[X_{[d]}\right] \\ & + \sum_{x_{[d]}\in\bigtimes_{k\in[d]}[m_k]\,:\,f\left[x_{[d]}\right]=0} \delta\left[Y_f^{\text{in}}, Y_f^{\text{out}}\right] \otimes \epsilon_{x_{[d]}}\left[X_{[d]}\right] . \end{aligned}$$

Example 2.5 (CNOT gate) The CNOT gate is the controlled Pauli-X, that is

$$\text{CNOT}\left[Y^{\text{in}}, Y^{\text{out}}, X\right] = \sigma_1\left[Y^{\text{in}}, Y^{\text{out}}\right] \otimes \epsilon_1\left[X\right] + \delta\left[Y^{\text{in}}, Y^{\text{out}}\right] \otimes \epsilon_0\left[X\right] .$$

For the identity function

$$\text{Id}(x) = x .$$

we have

$$C^{\text{Id}}\left[Y^{\text{in}}, Y^{\text{out}}, X\right] = \text{CNOT}\left[Y^{\text{in}}, Y^{\text{out}}, X\right] .$$

The CNOT is therefore the computation circuit to the identity function.

Functions with multiple output variables, i.e. $f : \bigtimes_{k\in[d]}[2] \to \bigtimes_{\ell\in[p]}[2]$, can be encoded image coordinate wise as a concatenation of the respective circuits.

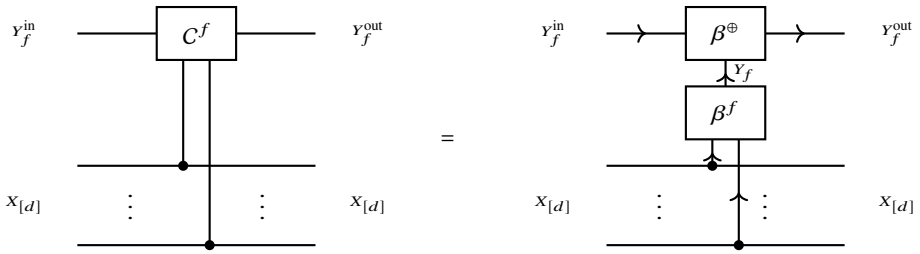


Fig. 2.7 Relation of computation circuit and basis encodings. The computation circuit of a function is the contraction of its basis encoding with the basis encoding of the mod2 sum $\oplus$.

2.3.1 Relation with basis encodings

Basis encodings are schemes to encode functions into directed boolean tensors, which enable reasoning about functions by tensor network contractions (see [11]). To connect with this formalism, let us now show the relation to the computation circuits.

Lemma 2.7 *For any boolean function we have (see Figure 2.7)*

$$C^f\left[Y_f^{\text{in}}, Y_f^{\text{out}}, X_{[d]}\right] = \left\langle \beta^f\left[Y_f, X_{[d]}\right], \beta^{\oplus}\left[Y_f^{\text{out}}, Y_f^{\text{in}}, Y_f\right]\right\rangle_{\left[Y_f^{\text{in}}, Y_f^{\text{out}}, X_{[d]}\right]} .$$

The computation circuit applied on the initial state

$$\sqrt{\frac{1}{2^d}}\mathbb{I}\left[X_{[d]}^{\text{out}}\right] \otimes \epsilon_0\left[Y^{\text{in}}\right]$$

prepares a q-sample to the normalized basis encoding.

In [11] (and more detailed in [10]) sparse decompositions of basis encodings have been studied. Based on the equivalence of computation circuits and contracted basis encodings, stated by Lem. 2.7 we transfer these decomposition schemes in the following.

Example 2.6 (Multiple-controlled NOT) Let us consider the d-ary controlled NOT gate

$$\text{MCNOT}^d\left[Y^{\text{in}}, Y^{\text{out}}, X_{[d]}\right] = \sigma_1\left[Y^{\text{in}}, Y^{\text{out}}\right] \otimes \epsilon_{1_{[d]}}\left[X_{[d]}\right] + \delta\left[Y^{\text{in}}, Y^{\text{out}}\right] \otimes \left(\mathbb{I}\left[X_{[d]}\right] - \epsilon_{1_{[d]}}\left[X_{[d]}\right]\right) .$$

This is the computation circuit of the d-ary logical conjunction

$$\wedge^d : [2]^d \to [2] \quad , \quad \wedge^d(x_{[d]}) = \begin{cases} 1 & \text{if} \quad x_{[d]} = 1_{[d]} \\ 0 & \text{else} \end{cases} .$$

Special instances are the CNOT for $d = 1$ (see Example 2.5) and the Toffoli gate for $d = 2$.

2.3.2 Circuit decomposition sparsity

We develop now a quantum equivalent to the circuit decompositions of basis encodings (see Sect. **??**) sis now a composition of circuits property, as stated in the next lemma. Similar to the introduction of hidden variables, we here utilize further working qubits to realize decompositions. To highlight the similarity, we call these sparsity principle circuit decomposition sparsity.

Lemma 2.8 *We have for functions* $q : \bigtimes_{k\in[d]}[2] \to \bigtimes_{\ell\in[p]}[2]$, $g : \bigtimes_{\ell\in[p]}[2] \to \bigtimes_{s\in[r]}[2]$ *(see Figure* **??***)*

$$C^{g\circ q}\left[Y^{\text{in}}_{g\circ q}, Y^{\text{out}}_{g\circ q}, X_{[d]}\right] \otimes \epsilon_0\left[Y^{\text{out}}_q\right]$$
$$= \left\langle \epsilon_0\left[Y^{\text{in}}_q\right], C^q\left[Y^{\text{in}}_q, Y^{\text{mid}}_q, X_{[d]}\right], C^g\left[Y_{\text{in},g\circ q}, Y^{\text{out}}_{g\circ q}, Y^{\text{mid}}_q\right], C^q\left[Y^{\text{mid}}_q, Y^{\text{out}}_q, X_{[d]}\right]\right\rangle_{\left[Y^{\text{in}}_{g\circ q}, Y^{\text{out}}_{g\circ q}, X_{[d]}, Y^{\text{out}}_q\right]}.$$

Here each copy of $Y_{g\circ q}$ denotes a tuple of r boolean variables and Y_q of p boolean variables.

Proof. We show the claim based on basis calculus and sketch the steps in Figure 2.8. In the first equation we use Lem. 2.7 and represent each computation circuit by a contraction of two basis encodings. In the second equation we use the identity

$$\left\langle \beta^{\oplus}\left[Y^{\text{out}}, Y^{\text{in}}, Y_q\right], \epsilon_0\left[Y^{\text{in}}\right]\right\rangle_{\left[Y^{\text{out}}, Y_q\right]} = \delta\left[Y^{\text{out}}, Y_q\right].$$

Now in the third equation we use that the contraction of basis encodings is the basis encoding of the composition function, and that $q \oplus q$ is the vanishing function, which basis encoding is a tensor product with $\epsilon_0\left[Y^{\text{out}}_q\right]$. We use we Lem. 2.7 again in the fourth equation to arrive at the claim. □

When having a syntactical decomposition of a propositional formula, we can iteratively apply the computation circuit decomposition theorem and prepare each connective by a circuit. We can decompose any propositional formula into logical connectives and prepare to each a computation circuit (exploiting further sparsity mechanisms). This works, when the target qubit of one connective is used as a value qubit of another.

This entailment of the working qubit with the distributed qubits can be resolved by an un-computation circuit, which is the adjoint of the computation circuit for the working qubits (see Figure 2.8). Note that while this is necessary in some applications, it is not necessary for the quantum rejection sampling presented here.

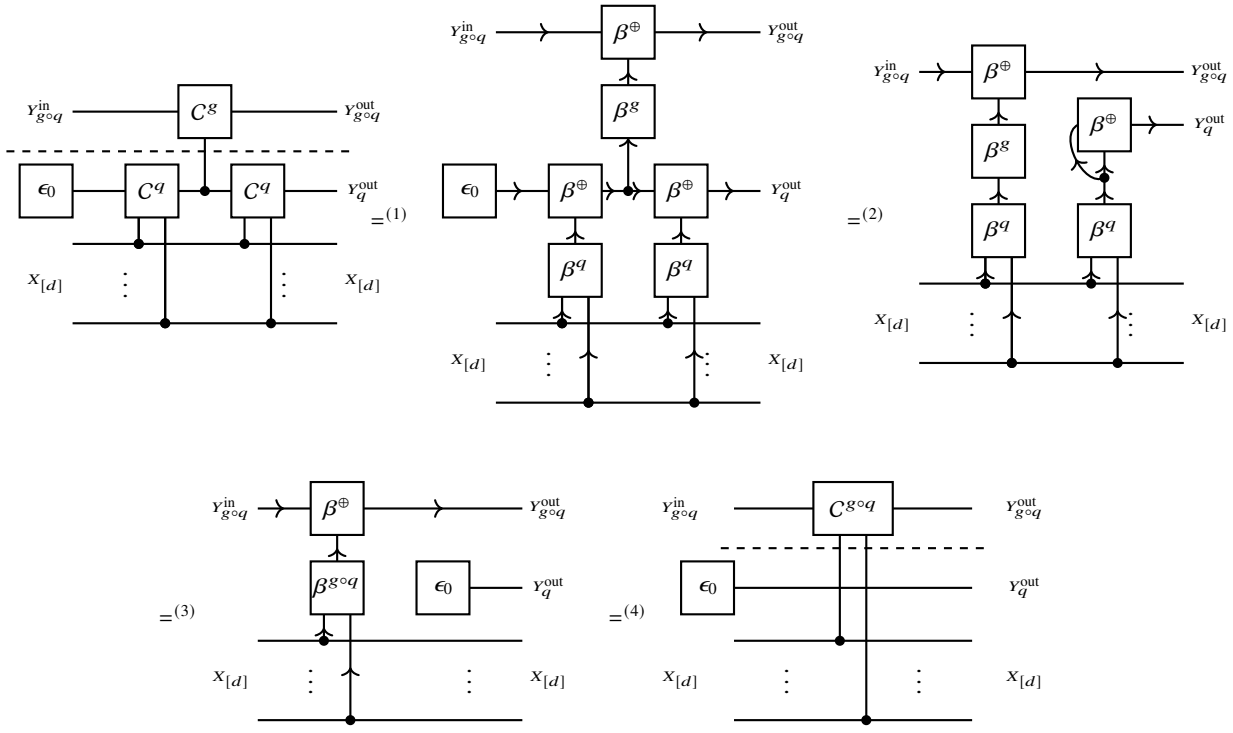


Fig. 2.8 Proof of Lem. 2.8 based on basis calculus.

Example 2.7 Let us reconsider the d-ary controlled NOT gate from Example 2.6. The d-ary logical conjunction is decomposed as a sequence of 2-ary logical conjunctions. The computation circuit of each 2-ary conjunction is a Toffoli gate applied to an input qubit and a worker qubit. Using Decomposition sparsity we therefore find the standard decomposition of d-ary controlled NOT gates into Toffoli gates.

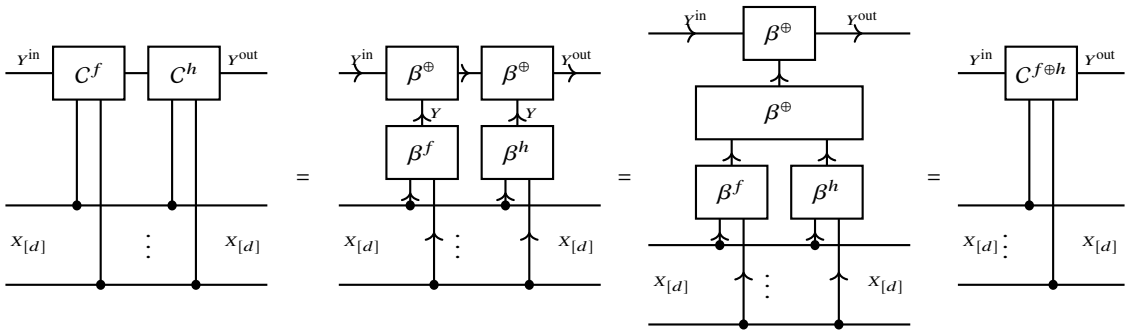


Fig. 2.9 Equivalence of the composition of computation circuits with the computation circuit of their mod2 sum (Lem. 2.9).

2.3.3 Circuit polynomial sparsity

While we have introduced additional auxiliary qubits to represent decomposable functions, we now investigate concatenations of computation circuits, which are sharing the same target qubit. As we will show, these are related to summations mod 2 (which are contractions in the semiring $\mathbb{F}_2$, see Sect. **??**), and can be constructed using Reed-Muller expansions (see Sect. **??**). To highlight the similarity to polynomial sparsity, which has been developed in Chapter **??**, we call this sparsity principle circuit polynomial sparsity.

Lemma 2.9 *Let* $\{f_\ell\left[X_{[d]}\right] : \ell \in [p]\}$ *be a set of propositional formulas, and let*

$$h\left[X_{[d]}\right] = \langle \sigma^f\left[X_{[d]}, L\right]\rangle^{\mathbb{F}_2}_{[X_{[d]}]}$$

be their summation mod 2*, which can be expressed using the selection encoding* $\sigma^f\left[X_{[d]}, L\right]$ *of the formula set. Then we have that*

$$\left\langle \{C^{f_\ell}\left[Y^\ell, Y^{\ell+1}, X_{[d]}\right] : \ell \in [p]\}\right\rangle_{\left[X_{[d]}\right]} = C^h\left[Y^0, Y^p, X_{[d]}\right] .$$

Proof. We first show the claim for $p = 2$, the sum of two formulas f_0 and f_1. We decompose in Figure 2.9 each computation circuit into a contraction of the functions basis encoding with the basis encoding of $\oplus$, the summation mod 2. By associativity of $\oplus$, we can rewrite the contraction using the basis encoding of $f_0 \oplus f_1$.

We then show the generic statement by induction over p. The statement is trivial for $p = 1$. Let us assume that the claim holds for $p - 1$ formulas, and let there be p formulas. By assumption, the concatenation of the first $p - 1$ computation circuits is the computation circuit of

$$\bigoplus_{k\in[p]} f_k$$

Applying the statement for $p = 2$, we conclude that the concatenation of all p computation circuits is the computation circuit of

$$h = \bigoplus_{k\in[p]} f_{\bar{k}} = \langle \sigma^f\left[X_{[d]}, L\right]\rangle^{\mathbb{F}_2}_{[X_{[d]}]} .$$

□

Recall that while the quantum circuit itself is defined in the complex sum-product semiring, we use the semiring $\mathbb{F}_2$ for function decomposition. The $\mathrm{CP}^{\mathbb{F}_2}$ decomposition is closely related to the Algebraic Normal Form of a Boolean polynomial. In the latter, we need to restrict to leg vectors $\mathbb{I}[X]$ and $\epsilon_1[X]$, that is forbid the usage of $\epsilon_0[X]$ vectors. The Algebraic Normal Form is also called positive-polarity Reed-Muller expansion. Allowing for the $\epsilon_0[X]$ vector in the $\mathrm{CP}^{\mathbb{F}_2}$ decomposition corresponds with mixed-polarity Reed-Muller (MPRM) expansions.

Let us now determine computation circuits for MPRM terms to get an encoding for arbitrary formulas in MPRM expansions.

Lemma 2.10 *A MPRM term $f_{(\mathcal{U},y_\mathcal{U})}$ has the computation circuit (see Figure 2.10)*

$$\begin{aligned}
&\left\langle \mathcal{V}^{f_{(\mathcal{U},y_\mathcal{U})}}\left[Y^{\text{in}}, Y^{\text{out}}, X_{[d]}\right], \delta\left[X_{[d]}, X^{\text{in}}_{[d]}, X^{\text{out}}_{[d]}\right]\right\rangle_{\left[Y^{\text{in}},Y^{\text{out}},X^{\text{in}}_{[d]},X^{\text{out}}_{[d]}\right]} \\
&\quad = \left\langle \{\text{MCNOT}^{|\mathcal{U}|}[Y^{\text{in}}, Y^{\text{out}}, X_\mathcal{U}]\} \cup \left(\bigcup_{k\in\mathcal{U},\, y_k=0} \{\sigma_1\left[X^{\text{in}}_k, X_k\right], \sigma_1\left[X_k, X^{\text{out}}_k\right]\}\right)\right. \\
&\qquad \left. \cup \left(\bigcup_{k\in\mathcal{U},\, y_k=1} \{\delta\left[X_k, X^{\text{in}}_k, X^{\text{out}}_k\right]\}\right)\right\rangle_{\left[Y^{\text{in}},Y^{\text{out}},X^{\text{in}}_{[d]},X^{\text{out}}_{[d]}\right]}.
\end{aligned}$$

The computation circuit of a formula f with MPRM expansion $\mathcal{M}$ is a concatenation of the computation circuits to $\mathcal{U} \in \mathcal{M}$.

Proof. For any $x^{\text{in}}_{[d]}, x^{\text{out}}_{[d]}$ the slices on both sides are vanishing, if $x^{\text{in}}_{[d]} \neq x^{\text{out}}_{[d]}$. If $x^{\text{in}}_{[d]} = x^{\text{out}}_{[d]}$ we have

$$\begin{aligned}
&\left\langle \{\text{MCNOT}^{|\mathcal{U}|}[Y^{\text{in}}, Y^{\text{out}}, X_\mathcal{U}]\} \cup \left(\bigcup_{k\in\mathcal{U},\, y_k=0} \{\sigma_1\left[X^{\text{in}}_k, X_k\right], \sigma_1\left[X_k, X^{\text{out}}_k\right]\}\right)\right. \\
&\qquad \left. \cup \left(\bigcup_{k\in\mathcal{U},\, y_k=1} \{\delta\left[X_k, X^{\text{in}}_k, X^{\text{out}}_k\right]\}\right)\right\rangle_{\left[Y^{\text{in}},Y^{\text{out}},X^{\text{in}}_{[d]}=x^{\text{in}}_{[d]},X^{\text{out}}_{[d]}=x^{\text{out}}_{[d]}\right]} \\
&= \left\langle \{\text{MCNOT}^{|\mathcal{U}|}[Y^{\text{in}}, Y^{\text{out}}, X_\mathcal{U}]\} \cup \{\epsilon_{(1-x^{\text{in}}_k)}\left[X_k\right] : k \in \mathcal{U},\, y_k = 0\} \cup \{\epsilon_{x^{\text{in}}_k}\left[X_k\right] : k \in \mathcal{U},\, y_k = 1\}\right\rangle_{\left[Y^{\text{in}},Y^{\text{out}}\right]} \\
&= \begin{cases} \text{NOT}[Y^{\text{in}}, Y^{\text{out}}] & \text{if} \quad f_{(\mathcal{U},y_\mathcal{U})}(x^{\text{in}}_{[d]}) = 1 \\ \delta\left[Y^{\text{in}}, Y^{\text{out}}\right] & \text{if} \quad f_{(\mathcal{U},y_\mathcal{U})}(x^{\text{in}}_{[d]}) = 0 \end{cases}.
\end{aligned}$$

Thus, the right contraction coincides for all slices on the variables $X^{\text{in}}_{[d]}$ and $X^{\text{out}}_{[d]}$ with the oracle of $f_{(\mathcal{U},y_\mathcal{U})}$. □

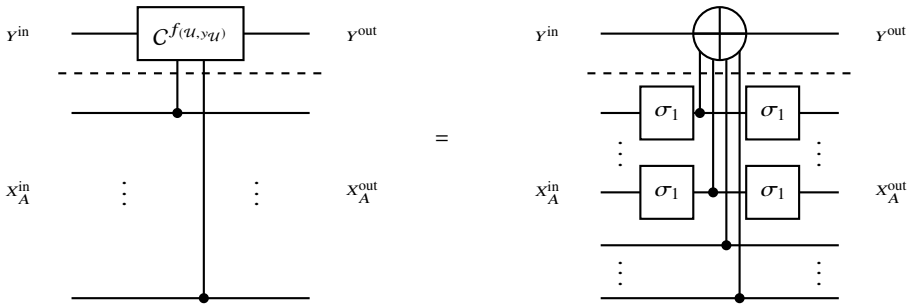


Fig. 2.10 Computation circuit to a MPRM term $f_{(\mathcal{U},y_\mathcal{U})}$, consisting in a multiple controlled NOT gate and Pauli-X gates for legs with $y_k = 0$.

Combining Lem. 2.10 and Lem. 2.9 we can create the computation circuit of any function based on the $\text{CP}^{\mathbb{F}_2}$ decomposition, which is equivalent to Reed-Muller expansions (see Sect. **??**).

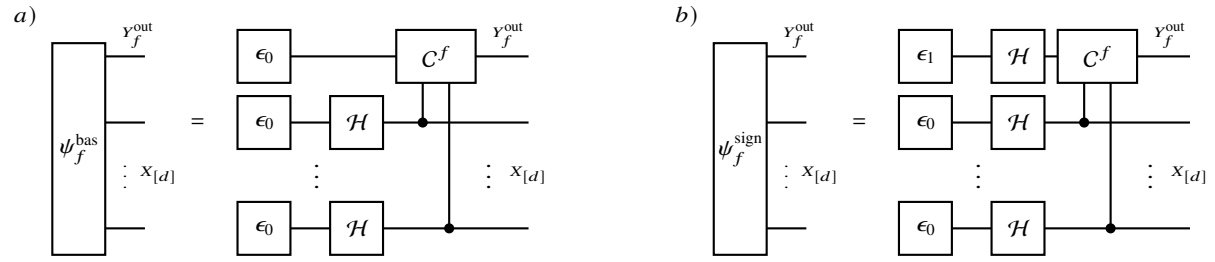


Fig. 2.11 Construction of the basis encoding state (a) and the sign encoding state (b) using the computation circuit of a function.

2.3.4 Function Encoding

We investigate how Boolean functions can be encoded in quantum states using computation circuits. Depending on the start state, we introduce two function encoding states.

Remark 2.1 (Basis and Amplitude Encodings in Quantum Computation [29] and tnreason*)* Basis Encoding in Quantum Computation refers to the representation of classical n bit strings by n qubit basis states, and is called one-hot encoding in tnreason. The Basis Encoding scheme in tnreason goes beyond this scheme and also encodes subsets by sums of one-hot encodings to the members of the set. In this way, relations and functions are represented by boolean tensors and contraction of them is refered to as Basis calculus.

Amplitude Encoding in Quantum Computation refers to the storage of complex numbers in the amplitudes of quantum states. The pendant in tnreason is the Coordinate Encoding scheme, where real numbers are stored in the coordinates of real-valued tensors. Compared to Amplitude Encoding, Coordinate Encoding does not have the normalization constraint of quantum states. The Amplitude Encoding of the square root of a probability distribution is sometimes called q-sample.

2.3.4.1 Basis encoding

Definition 2.5 The basis encoding of a boolean function f is the state

$$\psi_f^{\text{bas}}\left[X_{[d]}, Y\right] = \sqrt{\frac{1}{2^d}} \sum_{x_{[d]} \in \bigtimes_{k\in[d]}[2]} \epsilon_{x_{[d]}}\left[X_{[d]}\right] \otimes \epsilon_{f(x_{[d]})}\left[Y\right] .$$

Lemma 2.11 *When applying the computation circuit on the initial state*

$$\sqrt{\frac{1}{2^d}} \mathbb{I}\left[X_{[d]}\right] \otimes \epsilon_0\left[Y\right]$$

we get the basis encoding state (see Figure 2.11a).

Proof. Since the Pauli-X turns ϵ_0 into ϵ_1, and is applied exactly when $f(x_{[d]}) = 1$. □

Corollary 2.1 *Using Lem. 2.7 we have*

$$\psi_f^{\text{bas}}\left[X_{[d]}, Y\right] = \sqrt{\frac{1}{2^d}} \beta^f\left[X_{[d]}, Y\right]$$

The initial state can be prepared by applying Hadamard gates on all distributed variable qubits and preparing the head variable qubit in the ground state. The overlap of two basis encoding states (which share the same head variable) is proportional to the contraction of the corresponding formulas.

Lemma 2.12 *Let $f, \tilde{f}$ be two Boolean formulas on the states of $X_{[d]}$. Then we have*

$$\left\langle \psi_f^{\text{bas}}\left[X_{[d]}, Y\right], \psi_h^{\text{bas}}\left[X_{[d]}, Y\right] \right\rangle_{[\varnothing]} = \frac{1}{2^d} \left\langle f\left[X_{[d]}\right], \tilde{f}\left[X_{[d]}\right] \right\rangle_{[\varnothing]} .$$

Proof. We have

$$\begin{aligned}\left\langle \psi_f^{\text{bas}}\left[X_{[d]},Y\right],\psi_h^{\text{bas}}\left[X_{[d]},Y\right]\right\rangle_{[\varnothing]} &= \frac{1}{2^d}\sum_{x_{[d]}\in[2]^d}\left\langle \epsilon_{f\left[X_{[d]}=x_{[d]}\right]}\left[Y\right],\epsilon_{\tilde{f}\left[X_{[d]}=x_{[d]}\right]}\left[Y\right]\right\rangle_{[\varnothing]} \\ &= \frac{1}{2^d}\left\langle f\left[X_{[d]}\right],\tilde{f}\left[X_{[d]}\right]\right\rangle_{[\varnothing]} .\end{aligned}$$

□

2.3.4.2 Sign encoding

Definition 2.6 The sign encoding of a boolean function f is the state

$$\psi_f^{\text{sign}}\left[X_{[d]},Y\right] = \left(\sqrt{\frac{1}{2^d}}\sum_{x_{[d]}\in\bigtimes_{k\in[d]}[2]}(-1)^{f(x_{[d]})}\cdot\epsilon_{x_{[d]}}\left[X_{[d]}\right]\right)\otimes\sqrt{\frac{1}{2}}\left(\epsilon_0\left[Y\right]-\epsilon_1\left[Y\right]\right) .$$

Sign encodings can be prepared by the phase kickback trick, as we show next.

Lemma 2.13 (Phase kickback trick) *Applying the computation circuit* $C^f\left[Y^{\text{in}},Y^{\text{out}},X_{[d]}\right]$ *on a state*

$$\psi\left[X_{[d]}\right]\otimes\epsilon^-\left[A^{\text{in}}\right]$$

we get

$$\sum_{x_{[d]}\in\bigtimes_{k\in[d]}[m_k]}(-1)^{f(x_{[d]})}\psi\left[X_{[d]}=x_{[d]}\right]\cdot\epsilon_{x_{[d]}}\left[X_{[d]}\right]\otimes\epsilon^-\left[A^{\text{out}}\right] .$$

Proof. We get a disentangled ancilla qubit, because $\epsilon^-\left[A^{\text{in}}\right]$ is an eigenvector of the identity and the Pauli-X. For the identity, which is applied when $f(x_{[d]})=0$, the eigenvalue is 1. For the Pauli-X, which is applied when $f(x_{[d]})=0$, the eigenvalue is (-1). □

We get the sign encoding with the phase kickback trick, when we initialize with the q-sample of the uniform distribution.

Corollary 2.2 *When applying the computation circuit on the initial state*

$$\sqrt{\frac{1}{2^d}}\mathbb{I}\left[X_{[d]}\right]\otimes\sqrt{\frac{1}{2}}\left(\epsilon_0\left[Y\right]-\epsilon_1\left[Y\right]\right)$$

we get the sign encoding state (see Figure 2.11b). Note that the initial state is a Hadamard gate acting on the state $\epsilon_1\left[Y\right]$.

The sign encoding is also called the phase encoding. One can ask, whether we can use other ancilla states and get different phase kickbacks while keeping the ancilla disentangled. However, since we construct here computation circuit we are restricted to the eigenvectors of the Pauli-X gate, which is the Hadamard basis

$$\epsilon^-\left[Y\right]=\sqrt{\frac{1}{2}}\left(\epsilon_0\left[Y\right]-\epsilon_1\left[Y\right]\right) \quad , \quad \epsilon^+\left[Y\right]=\sqrt{\frac{1}{2}}\left(\epsilon_0\left[Y\right]+\epsilon_1\left[Y\right]\right)$$

with eigenvalues (-1) and 0.

The initial state can be prepared by applying Hadamard gates on all distributed variable qubits and preparing the head variable qubit by applying a Hadamard gate on the first basis state, i.e.

$$\frac{1}{\sqrt{2}}\left(\epsilon_1\left[Y_{\text{out}}\right]-\epsilon_0\left[Y_{\text{out}}\right]\right)=\mathcal{H}\left[Y_{\text{out}},Y_{\text{in}}\right]\epsilon_0\left[Y_{\text{in}}\right] ,$$

Lemma 2.14 *Let $f, \tilde{f}$ be two Boolean formulas on the states of $X_{[d]}$. Then we have*

$$\left\langle \psi_f^{\text{sign}}\left[X_{[d]}, Y\right], \psi_h^{\text{sign}}\left[X_{[d]}, Y\right]\right\rangle_{[\varnothing]} = \frac{1}{2^d}\left(\left\langle f\left[X_{[d]}\right], \tilde{f}\left[X_{[d]}\right]\right\rangle_{[\varnothing]} - \left\langle f\left[X_{[d]}\right], \neg\tilde{f}\left[X_{[d]}\right]\right\rangle_{[\varnothing]}\right).$$

Proof. We have

$$\begin{aligned}\left\langle \psi_f^{\text{sign}}\left[X_{[d]}, Y\right], \psi_h^{\text{sign}}\left[X_{[d]}, Y\right]\right\rangle_{[\varnothing]} &= \frac{1}{2^d}\sum_{x_{[d]}\in[2]^d}(-1)^{f\left[X_{[d]}=x_{[d]}\right]+\tilde{f}\left[X_{[d]}=x_{[d]}\right]} \\ &= \frac{1}{2^d}\sum_{x_{[d]}\in[2]^d : f\left[X_{[d]}=x_{[d]}\right]=\tilde{f}\left[X_{[d]}=x_{[d]}\right]} 1 + \sum_{x_{[d]}\in[2]^d : f\left[X_{[d]}=x_{[d]}\right]\neq\tilde{f}\left[X_{[d]}=x_{[d]}\right]} (-1) \\ &= \frac{1}{2^d}\left(\left\langle f\left[X_{[d]}\right], \tilde{f}\left[X_{[d]}\right]\right\rangle_{[\varnothing]} - \left\langle f\left[X_{[d]}\right], \neg\tilde{f}\left[X_{[d]}\right]\right\rangle_{[\varnothing]}\right).\end{aligned}$$

□

Limitations of sign encodings:

- The prepared state ψ_f^{sign} is distinguished from the state $\psi_{\neg f}^{\text{sign}}$ only by a non-observable phase factor

$$\psi_f^{\text{sign}} = (-1)\cdot\psi_{\neg f}^{\text{sign}}$$

- With this construction the statistic qubit is disentangled with the distributed qubits (in contrast to basis encoding states). Further manipulation of the statistic qubits alone (as we do with activation circuits) will not retrieve any information about the encoded function.
- When doing the sign encoding procedure for multiple formulas, we get a disentangled state of the head variables with the distributed variables encoded in the sign encoding state of $\bigoplus_{f\in\mathcal{F}} f$ (where by $\bigoplus$ we denote the mod2 sum of boolean functions).

2.4 Quantum algorithms

We now review standard Quantum algorithms in the tensor network formalism.

2.4.1 Quantum Fourier transform

Recall the tensor network decomposition of the Fourier transform developed in Sect. 1.2.

When $V[X_{[d]}]$ is a quantum state, the Fourier transform can be understood as a unitary evolving the state. Since both the controlled U tensor and the Hadamard H are unitary, each of them can be implemented by a sequence of gates. For sparse implementation one can exploit that U is a tensor product of matrices (i.e. single qubit gates).

We introduce the gates

$$R^k[X^{\text{in}}, X^{\text{out}}] = \begin{pmatrix}1 & 0\\ 0 & \exp\left[2\pi\mathrm{i}\frac{1}{2^k}\right]\end{pmatrix}\left[X^{\text{in}}, X^{\text{out}}\right].$$

The unitary U^d in the tensor network decomposition of the fourier transform (see Sect. 1.2.2) is then the tensor product

$$U^d[X_{\text{in}}^{[d]}, X_{\text{out}}^{[d]}] = \bigotimes_{k\in[d]} R^{k+2}[X_{\text{in}}^k, X_{\text{out}}^k].$$

Controlling U^d by X_d is equivalent to controlling each R^k gate.

To realize the QFT we need d single-qubit Hadamard gates and $\sum_{k\in[d]} k = \frac{d\cdot(d-1)}{2}$ controlled diagonal single-qubit gates (to realize the U^k transforms). The gate complexity is therefore $O(d^2)$.

2.4.1.1 Walsh-Hadamard transform

In case of $m_k = 2$, the Fourier transform is known to be the Walsh-Hadamard tranform. The transform is performed by applying Hadamard gates on each variable. This is equivalent with contraction of the tensor

$$\mathcal{H}^d\left[X_{[d],\text{in}}, X_{[d],\text{out}}\right] = \bigotimes_{k\in[d]} \mathcal{H}\left[X_k^{\text{in}}, X_k^{\text{out}}\right] ,$$

which coordinates are

$$\mathcal{H}^d\left[X_{[d],\text{in}} = x_{[d],\text{in}}, X_{[d],\text{out}} = x_{[d],\text{out}}\right] = \sqrt{\frac{1}{2^d}}\,(-1)^{\sum_{k\in[d]} x_{k,\text{in}}\cdot x_{k,\text{out}}} .$$

We have

$$\left\langle \epsilon_0\left[X^{\text{in}}\right], \mathcal{H}\left[X^{\text{in}}, X^{\text{out}}\right]\right\rangle_{[X^{\text{out}}]} = \sqrt{\frac{1}{2}}\mathbb{I}\left[X^{\text{out}}\right]$$
$$\left\langle \epsilon_1\left[X^{\text{in}}\right], \mathcal{H}\left[X^{\text{in}}, X^{\text{out}}\right]\right\rangle_{[X^{\text{out}}]} = \sqrt{\frac{1}{2}}\begin{pmatrix}1\\-1\end{pmatrix}\left[X^{\text{out}}\right]$$

and conversely

$$\left\langle \sqrt{\frac{1}{2}}\mathbb{I}\left[X^{\text{in}}\right], \mathcal{H}\left[X^{\text{in}}, X^{\text{out}}\right]\right\rangle_{[X^{\text{out}}]} = \epsilon_0\left[X^{\text{out}}\right]$$
$$\left\langle \sqrt{\frac{1}{2}}\begin{pmatrix}1\\-1\end{pmatrix}\left[X^{\text{in}}\right], \mathcal{H}\left[X^{\text{in}}, X^{\text{out}}\right]\right\rangle_{[X^{\text{out}}]} = \epsilon_1\left[X^{\text{out}}\right] .$$

If and only if the function is constant, then the transformed state is the ground state.

Let us present two known algorithms as the measurement of states prepared by computation circuits.

Example 2.8 (Deutsch-Jozsa Test) We measure the probability of the ground state of the sign encoding after a Walsh-Hadamard transform. The probability of measuring the ground state after a Walsh-Hadamard transform of the distributed qubits is then

$$\frac{1}{2^d}\cdot\left\langle\left|\left\langle\psi_f^{\text{sign}}\left[X_{[d]}, Y\right]\right\rangle_{[Y]}\right|^2\right\rangle_{[\varnothing]} = \frac{1}{2^{2\cdot d}}\left(\langle f\rangle_{[\varnothing]} - \langle\neg f\rangle_{[\varnothing]}\right)^2 .$$

Thus, if and only if the probability of the ground state is 1, we have that the formula is constant, that is $f \in \{\top, \bot\}$. If and only if the probability of the ground state is 0, we have that the formula is balanced, that is $\langle f\rangle_{[\varnothing]} = \langle\neg f\rangle_{[\varnothing]}$. When its known, that the formula is either constant or balanced, we can decide which one it is by measuring the probability of the ground state. This is the Deutsch-Josza algorithm [8].

Note that when allowing for an exponentially small error the decision can be efficiently performed by a classical algorithm. When drawing n samples and deciding on $\mu = 0.5$ if and only if in the samples a 0 and a 1 is observed, we have an error probability of 0 when $\mu \in \{0, 1\}$ and of

$$\mathbb{P}^{E|\mu=0.5} = 2\cdot\frac{1}{2^n} .$$

The quantum advantage is thus bound to the demand of an error-free decision, for which a generic classical algorithm would iterate through all states and is therefore exponentially slower than the quantum alternative.

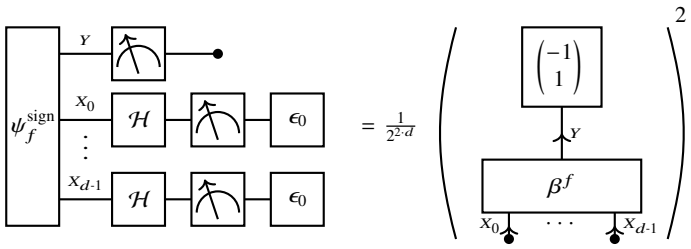


Fig. 2.12 Measurement setup for the Deutsch-Jozsa algorithm on the formula f.

Another generalizations would be period detection. Deutsch-Jozsa can be generalized to estimating the moments of a distribution given its q-sample. Note, that we have to demand a q-sample (i.e. vanishing phase tensor), otherwise the measurement is affected by the phase tensor and the Deutsch-Jozsa test is effectively done wrt another function. If for example the coordinates of the phase tensor are multiples of π, and the phases are thus $+1/-1$, then the test is done wrt the function $f \oplus (\frac{1}{\pi}\phi)$ instead of f.

Example 2.9 (Inversion Test) Concatenating the computation circuits to two formulas f and h prepares the state

$$\psi^{\text{bas}}_{f\oplus h}\left[X_{[d]}, Y\right]$$

The probability of measuring the ground state after a Walsh-Hadamard transform of the distributed qubits (see Figure 2.13) is then

$$\begin{aligned}\frac{1}{2^d}\left|\left\langle \psi^{\text{bas}}_{f\oplus h}\left[X_{[d]}, Y\right]\right\rangle_{[Y]}\right|^2 &= \frac{1}{2^{2\cdot d}}\left[\langle f\oplus h\rangle_{[\varnothing]}\left\langle \neg(f\oplus h)\right\rangle_{[\varnothing]}\right] \\ &= \frac{1}{2^{2\cdot d}}\begin{bmatrix}\#\left\{x_{[d]} \in \bigtimes_{k\in[d]}[2] \,:\, f\left[X_{[d]}=x_{[d]}\right] = h\left[X_{[d]}=x_{[d]}\right]\right\} \\ \#\left\{x_{[d]} \in \bigtimes_{k\in[d]}[2] \,:\, f\left[X_{[d]}=x_{[d]}\right] \neq h\left[X_{[d]}=x_{[d]}\right]\right\}\end{bmatrix}.\end{aligned}$$

The second equation holds since $\neg(f \oplus h)$ is equivalent with the biconditional $f \Leftrightarrow h$.

This is in more generality the inversion test of measuring the overlap of ψ^{bas}_f with ψ^{bas}_h (see [29]). Here we used that the computation circuit is self-adjoint and understand the concatenation as the unitaries preparing ψ^{bas}_f and the adjoint (together with the Walsh-Hadamard transform) of ψ^{bas}_h.

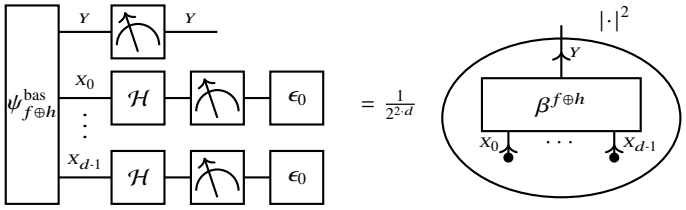


Fig. 2.13 Measurement setup for the inversion test on the formulas f and h. We measure the probability of the statistic qubit Y, when the distributed qubits $X_{[d]}$ are in the ground state after a Walsh-Hadamard transform. This is equal to a scaled and squared contraction of the basis encoding to the formula $f \oplus h$.

Quantum Circuits such as the SWAP test and the Hadamard test ([29, Section 3.6.1]) can be used to measure overlaps of quantum states, which are the squared absolutes of contractions of two state tensors.

- When we have preparation schemes for two tensors, we can control them with a common ancilla qubit and measure their contraction by a Hadamard test (alternatively, using the SWAP test and state preparation in two registers).
- Can we extend these schemes to contractions of more general tensor networks?

While the preparation of contracted basis encodings is done efficiently, accessing the prepared state by measurements is a bottleneck. This is a typical *quantum state tomography* problem [24]. Sparsity of the contracted state can be exploited to probabilistically guarantee precise estimates [12]. However, when deciding whether f is satisfiable, based on the prepared contraction $\sqrt{\left\langle \beta^f \left[Y, X_{[d]}\right]\right\rangle_{[Y|\varnothing]}}$, we have an exponentially small error tolerance. Moreover, the applied sample mean is the UMVUE (Uniformly Minimum Variance Unbiased Estimator, see [3]), and can thus not be improved by other unbiased estimators.

2.4.1.2 Period detection

QFT is a typical component in Quantum algorithms: QFT can be done efficiently with $\wr(d^2)$ controlled single-qubit gates. If the Fourier transform is a basis vector, a single measurement is sufficient to determine it. We now investigate how this can be used in the detection of periods.

Based on the support of the Fourier transform, one can derive the periods of a function. Periods are indices $(Z_0, \ldots, Z_{d-1})$ such that

$$\tau\left[X_{[d]} = x_{[d]}\right] = \tau\left[X_{[d]} = x_{[d]} \oplus z_{[d]}\right]$$

where $\oplus$ is the position-wise sum mod m_k.

Each supported index of the Fourier transform corresponds with periods in the function. The periods of the functions are thus at least the intersection of those (when assuming more structure one might have more periods).

In case of the Walsh-Hadamard transform (i.e. $m_k = 2$), we can easily determine the periods corresponding to each basis vector. The inverse Fourier transform of $\epsilon_{\ell_{[d]}}\left[L_{[d]}\right]$ is the tensor

$$\left(\bigotimes_{k\in[d]\,:\,\ell_k=0} \epsilon^+\left[X_k\right]\right) \otimes \left(\bigotimes_{k\in[d]\,:\,\ell_k=1} \epsilon^-\left[X_k\right]\right).$$

For each $\epsilon^+\left[X_k\right]$ we have a period generator of ϵ_k. For each pair of $\epsilon^-\left[X_k\right] \otimes \epsilon^-\left[X_{\tilde{k}}\right]$ we have a period generator of $\epsilon_k + \epsilon_{\tilde{k}}$.

One example is the Deutsch-Jozsa test, where the transformed is the $\epsilon_0\left[L_{[d]}\right]$ state, if and only if the function is constant. In this case, any vector is a period, since the group itself is generated by the period generators.

2.4.1.3 Generic Hidden Subgroup Problems

Based on the ideas in Sect. 1.2.4.5 we now apply the QFT routine to solve hidden subgroup problems. The main strategy is the support estimation of the Fourier transformed basis encoding state of a function, which is the annihilator of the hidden subgroup. Note that it suffices to get a set of linear independent basis states, which cardinality is the same as the minimal number of generators, to get the annihilator. The circuit is shown in Figure 2.14.

Example 2.10 (Simons problem) Simons problem is the HSP in case of a generic hidden subgroup H in $G = \bigtimes_{k\in[d]}[2]$ with rank n (the minimal number of generators). The annihilator then has rank $d - n$ and any set of $d - n$ many linear independent vectors spans it (using that $m_k = 2$). Given a set of measurements $z^j_{[d]}$ for $j \in [d - n]$, which are linearly independent, the subgroup H is constructed by the solution set of the equation system

$$\forall j \in [d-n] : \left\langle z^j_{[d]}, x_{[d]} \right\rangle_{[\varnothing]} = 0 \mod 2\,.$$

2.4.2 Quantum Phase Estimation

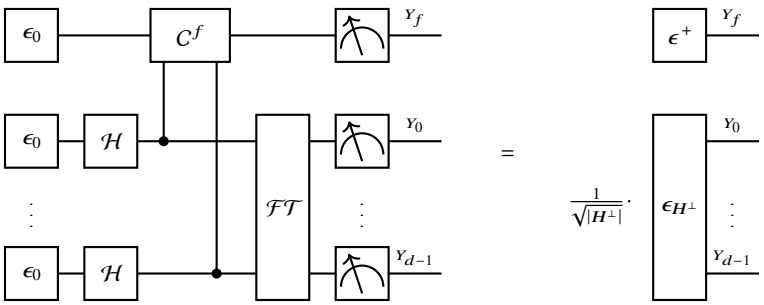


Fig. 2.14 Measurement setup to solve the Hidden Subgroup Problem. The basis encoding state of a formula f is prepared and the Fourier transform performed on the incoming qubits $X_{[d]}$. If f satisfies the promise to a subgroup H, then the measurement distribution is the uniform distribution of the annihilator elements.

When having a circuit implementing a unitary, the phase of the eigenvalues can be estimated using the Quantum Phase Estimation routine [17, 6].

Let $\mathcal{U}\left[X_{\text{in}}^{[d]}, X_{\text{out}}^{[d]}\right]$ be a unitary then the absolute of any eigenvalue is 1 and the eigenvalue is expressible by

$$\exp\left[2\pi\mathrm{i}\cdot\phi\right]$$

for some $\phi \in [0,1]$. Let $\phi = 0.\phi_0\cdots\phi_{n-1}$ be a 2-adic representation of ϕ, that is

$$\phi = \sum_{k\in[n]} 2^{-k-1}\phi_k\,.$$

In the quantum phase estimation routine the Fourier transform of $\epsilon_{\phi_0,\ldots,\phi_{n-1}}\left[L_{[n]}\right]$

$$\sqrt{\frac{1}{2^n}}\cdot\bigotimes_{k\in[n]}\left(\epsilon_0\left[L_0\right] + \exp\left[2\pi\mathrm{i}\cdot 0.\phi_{n-k-1}\cdots\phi_{n-1}\right]\right)$$

is prepared as explained below. The inverse Fourier transform on the register $L_{[n]}$ results then in the state

$$\epsilon_{\phi_0,\ldots,\phi_{n-1}}\left[L_{[n]}\right]\otimes\psi\left[X_{[d]}\right]$$

and ϕ_k for $k\in[n]$ can be retrieved by a single computational basis measurement of the register $L_{[n]}$.

To prepare the Fourier transform of $\epsilon_{\phi_0,\ldots,\phi_{n-1}}\left[X_{[d]}\right]$, we start with the state

$$\epsilon^+\left[L\right]\otimes\psi\left[X_{[d]}\right]\,.$$

When applying $\mathcal{U}\left[X_{\text{in}}^{[d]}, X_{\text{out}}^{[d]}\right]$ 2^k times controlled on L we get the state

$$\sqrt{\frac{1}{2}}\left(\epsilon_0\left[L_0\right] + \exp\left[2\pi\mathrm{i}\cdot 0.\phi_{n-k-1}\cdots\phi_{n-1}\right]\right)\otimes\psi\left[X_{[d]}\right]$$

Now we apply for each $k\in[n]$ the unitary $\mathcal{U}\left[X_{\text{in}}^{[d]}, X_{\text{out}}^{[d]}\right]$ 2^k times controlled on L_k and get

$$\sqrt{\frac{1}{2^n}}\mathbb{I}\left[L_{[n]}\right]\otimes\psi\left[X_{[d]}\right]\,.$$

2.4.3 Grover operator

We derive quantum inference schemes based on the Grover operator, which we want to first introduce in full generality.

2.4.3.1 Definition

Let us assume we have two orthogonal quantum states $\psi^{\parallel}\left[X_{[d]}\right], \psi^{\perp}\left[X_{[d]}\right]$ spanning a subspace $V = \text{span}\left(\psi^{\parallel}\left[X_{[d]}\right], \psi^{\parallel}\left[X_{[d]}\right]\right)$ and let us choose another state

$$\psi^{0}\left[X_{[d]}\right] \in V \, .$$

Then there is $\alpha \in [0, 4\pi)$ with

$$\psi^{0}\left[X_{[d]}\right] = \sin\left(\frac{\alpha}{2}\right)\psi^{\parallel}\left[X_{[d]}\right] + \cos\left(\frac{\alpha}{2}\right)\psi^{\perp}\left[X_{[d]}\right] \, .$$

Consider the reflection about the states $\psi^{\perp}$ and ψ^{0} by

$$\begin{aligned}
\mathcal{S}^{\perp}\left[X^{\text{in}}_{[d]}, X^{\text{out}}_{[d]}\right] &= 2 \cdot \psi^{\perp}\left[X^{\text{in}}_{[d]}\right] \otimes \psi^{\perp}\left[X^{\text{out}}_{[d]}\right] - \delta\left[X^{\text{in}}_{[d]}, X^{\text{out}}_{[d]}\right] \\
\mathcal{S}^{0}\left[X^{\text{in}}_{[d]}, X^{\text{out}}_{[d]}\right] &= 2 \cdot \psi^{0}\left[X^{\text{in}}_{[d]}\right] \otimes \psi^{0}\left[X^{\text{out}}_{[d]}\right] - \delta\left[X^{\text{in}}_{[d]}, X^{\text{out}}_{[d]}\right] \, ,
\end{aligned}$$

and their concatenation

$$G\left[X^{\text{in}}_{[d]}, X^{\text{out}}_{[d]}\right] = \left\langle \mathcal{S}^{\perp}\left[X^{\text{in}}_{[d]}, X^{\text{mid}}_{[d]}\right], \mathcal{S}^{0}\left[X^{\text{mid}}_{[d]}, X^{\text{out}}_{[d]}\right]\right\rangle_{\left[X^{\text{in}}_{[d]}, X^{\text{out}}_{[d]}\right]}$$

which is called the *Grover operator*.

2.4.3.2 Action on projected quantum states

The Grover operator leaves any quantum state ψ invariant, which is orthogonal to $\psi^{\parallel}$ and $\psi^{\perp}$ (such states get a factor (-1) by both reflection operations). The projection of the quantum state onto V gets rotated by the Grover operator (see Figure 2.15). To be more precise, we denote the projection by $(\mathcal{P}\psi)[L]$ which has the coordinates

$$\begin{aligned}
(\mathcal{P}_{V}\psi)\left[L = 0\right] &= \left\langle \left(\psi\left[X_{[d]}\right]\right)^{\dagger}, \psi^{\parallel}\left[X_{[d]}\right]\right\rangle_{[\varnothing]} \\
(\mathcal{P}_{V}\psi)\left[L = 1\right] &= \left\langle \left(\psi\left[X_{[d]}\right]\right)^{\dagger}, \psi^{\parallel}\left[X_{[d]}\right]\right\rangle_{[\varnothing]} \, .
\end{aligned}$$

We define a matrix representation of the Grover operator as

$$G^{\text{red}}\left[L^{\text{out}}, L^{\text{in}}\right] = \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix}\left[L^{\text{out}}, L^{\text{in}}\right]$$

and it holds for any quantum state $\psi\left[X_{[d]}\right]$ that

$$(\mathcal{P}_{V}G\psi)\left[L^{\text{out}}\right] = \left\langle G^{\text{red}}\left[L^{\text{out}}, L^{\text{in}}\right], (\mathcal{P}_{V}\psi)\left[L^{\text{in}}\right]\right\rangle_{[\text{out}]} \, .$$

The Grover operator therefore has the eigenvalues:

- $\exp\left[\pm i\alpha\right]$ with the eigen spaces spanned by $\psi^{\parallel} \pm i\psi^{\perp}$
- 1 with the eigen space by the orthogonal complement of V

Application the Grover operator $j \in \mathbb{N}$ times on ψ^{0} results in the state

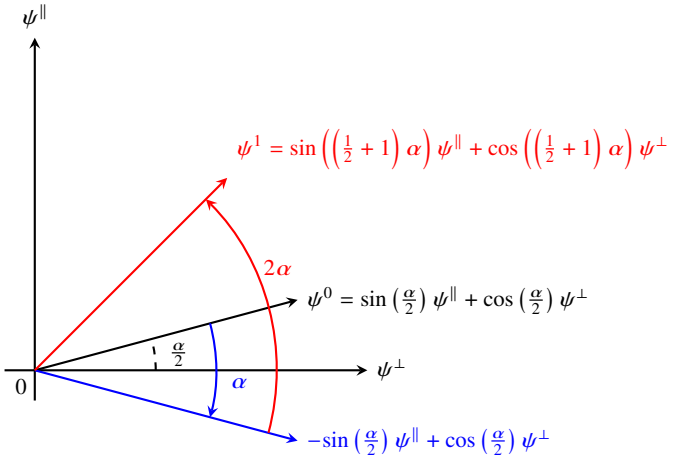


Fig. 2.15 Grover operator as a rotation in span $(\psi^{\parallel}, \psi^{\perp})$.

$$\psi^j\left[X_{[d]}\right] = \sin\left(\left(\frac{1}{2}+j\right)\alpha\right)\psi^{\parallel}\left[X_{[d]}\right] + \cos\left(\left(\frac{1}{2}+j\right)\alpha\right)\psi^{\perp}\left[X_{[d]}\right] .$$

2.4.3.3 Angle measurement

Using Quantum Phase Estimation we estimate the angle of the Grover rotation.

The eigenvalues of the Grover operator are (see Sect. 2.4.3) $\exp[\pm \mathrm{i}\alpha]$ and 1, and have therefore phases $\frac{\pm\alpha}{2\pi\mathrm{i}}$ and 0. Note that span $(\psi^{\parallel}, \psi^{\perp})$ is also spanned by the first two eigenvectors. When the state of the register is in span $(\psi^{\parallel}, \psi^{\perp})$ we therefore measure either $\frac{\alpha}{2\pi\mathrm{i}}$ or $\frac{2\pi\mathrm{i}-\alpha}{2\pi\mathrm{i}}$.

2.4.3.4 Circuit Realization of the reflection operators

Let us now investigate how the reflection operations in the Grover operator can be realized by circuits.

First, the reflection about the state $\epsilon_{1_{[d]}}\left[X_{[d]}\right]$ is realized by a multiple controlled Pauli-Z (see Figure 2.16)

$$-\mathcal{S}^{\epsilon_{1_{[d]}}}\left[X^{\mathrm{in}}_{[d]}, X^{\mathrm{out}}_{[d]}\right] = \left\langle \mathrm{MC}^{\mathrm{d}-1}\sigma_3\left[X^{\mathrm{in}}_{d-1}, X^{\mathrm{out}}_{d-1}, X_{[d-1]}\right], \delta\left[X'_{[d-1]} X^{\mathrm{in}}_{[d-1]}, X^{\mathrm{out}}_{[d-1]}\right]\right\rangle_{\left[X^{\mathrm{in}}_{[d]}, X^{\mathrm{out}}_{[d]}\right]}$$

Note that a global phase of -1 is introduced, which leaves the measurement distribution invariant. Adding layers of Pauli-X gates to each $k \in [d]$ before and after $-\mathcal{S}^{\epsilon_{1_{[d]}}}\left[X^{\mathrm{in}}_{[d]}, X^{\mathrm{out}}_{[d]}\right]$ (see Figure 2.16) prepares the negated reflection $-\mathcal{S}^{\epsilon_{0_{[d]}}}\left[X^{\mathrm{in}}_{[d]}, X^{\mathrm{out}}_{[d]}\right]$, since

$$\epsilon_1\left[X^{\mathrm{out}}_k\right] = \left\langle \sigma_3\left[X^{\mathrm{out}}_k, X^{\mathrm{in}}_k\right], \epsilon_0\left[X^{\mathrm{in}}_k\right]\right\rangle_{\left[X^{\mathrm{out}}_k\right]} .$$

Having a unitary $\mathcal{U}$ preparing ψ by action on the ground state, we realize the reflection (see Figure 2.16)

$$\mathcal{S}^{\psi}\left[X^{\mathrm{in}}_{[d]}, X^{\mathrm{out}}_{[d]}\right] = \left\langle \mathcal{U}^{\dagger}\left[X^{\mathrm{in}}_{[d]}, X^{\mathrm{mid},0}_{[d]}\right], \mathcal{S}^{\epsilon_{0_{[d]}}}\left[X^{\mathrm{mid},0}_{[d]}, X^{\mathrm{mid},1}_{[d]}\right], \mathcal{U}\left[X^{\mathrm{mid},1}_{[d]}, X^{\mathrm{out}}_{[d]}\right]\right\rangle_{\left[X^{\mathrm{in}}_{[d]}, X^{\mathrm{out}}_{[d]}\right]} .$$

Alternatively we can use the phasekick technique (Lem. 2.13) when acting on an disentangled ancilla qubit prepared in the Hadamard basis state

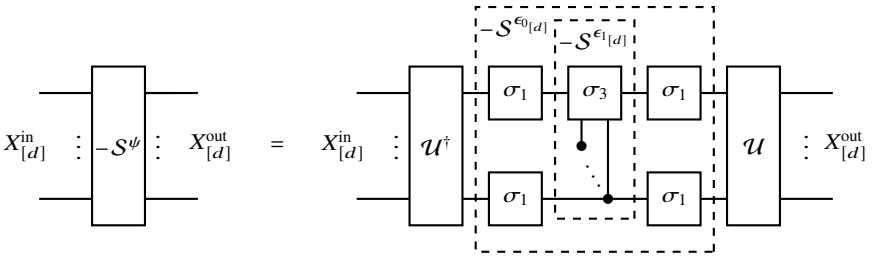


Fig. 2.16 Preparation of the negated reflection operator $-\mathcal{S}^{\psi}$ about a state ψ, which is prepared by a unitary $\mathcal{U}$ acting on the ground state $\epsilon_{0_{[d]}}\left[X_{[d]}\right]$. The multiply controlled Pauli-Z is the negated reflection $-\mathcal{S}^{1_{[d]}}$, and extended to the negated reflection $-\mathcal{S}^{1_{[d]}}$ about the ground state by adding layers of single qubit Pauli-X. When concatenating with $\mathcal{U}^{\dagger}$ and $\mathcal{U}$, the $-\mathcal{S}^{\psi}$ results.

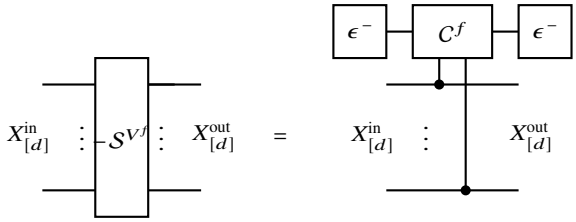


Fig. 2.17 Realization of the reflection across the subspace of one-hot encoded models to a Boolean function f. An ancilla qubit is prepared in the Hadamard basis state ϵ^- and serves as the head qubit of the oracle C^f. By the phase kickback effect, the qubit stays disentangled and the reflection $-\mathcal{S}^{V^f}$ is effectively performed on the qubits $X_{[d]}$.

$$\epsilon^-\left[Y\right] = \frac{1}{\sqrt{2}} \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}\left[Y\right] .$$

We notice, that

$$\left\langle C^f\left[Y^{\text{out}}, Y^{\text{in}}, X_{[d]} = x_{[d]}\right], \epsilon^-\left[Y^{\text{in}}\right]\right\rangle_{[Y^{\text{out}}]} = \begin{cases} \epsilon^-\left[Y^{\text{in}}\right] & \text{if} \quad f(x_{[d]}) = 0 \\ -\epsilon^-\left[Y^{\text{in}}\right] & \text{if} \quad f(x_{[d]}) = 1 \end{cases}$$

Let us consider the span of one-hot encoded models of $\neg f$, namely

$$V^{\neg f} := \text{span}\left(\epsilon_{x_{[d]}}\left[X_{[d]}\right] \ : \ f(x_{[d]}) = 0\right)$$

and the span of one-hot encoded models

$$V^{f} := \text{span}\left(\epsilon_{x_{[d]}}\left[X_{[d]}\right] \ : \ f(x_{[d]}) = 1\right) .$$

The reflection across $V^{\neg f}$ is then performed by (see Figure 2.17)

$$\mathcal{S}^{V^{\neg f}}\left[X^{\text{out}}_{[d]}, X^{\text{in}}_{[d]}\right] = \left\langle \epsilon^-\left[Y^{\text{out}}\right], C^f\left[Y^{\text{out}}, Y^{\text{in}}, X_{[d]}\right], \epsilon^-\left[Y^{\text{in}}\right], \delta\left[X_{[d]}, X^{\text{out}}_{[d]}, X^{\text{in}}_{[d]}\right]\right\rangle_{\left[X^{\text{out}}_{[d]}, X^{\text{in}}_{[d]}\right]}$$

We further have

$$\mathcal{S}^{V^f}\left[X^{\text{out}}_{[d]}, X^{\text{in}}_{[d]}\right] = -\mathcal{S}^{(V^f)^{\perp}}\left[X^{\text{out}}_{[d]}, X^{\text{in}}_{[d]}\right] = -\mathcal{S}^{V^{\neg f}}\left[X^{\text{out}}_{[d]}, X^{\text{in}}_{[d]}\right] .$$

When preparing a disentangled ancilla qubit Y in the basis state $\epsilon^-\left[Y\right]$, the oracle C^f is effectively a negated reflection across the span of one-hot encoded models of f.

2.4.4 Moment Estimation

As an example, we here investigate how to use the introduced quantum routines to measure the mean of a function given a probability distribution. Given a distribution $\mathbb{P}\left[X_{[d]}\right]$, the mean of a propositional formula $f\left[X_{[d]}\right]$ is the contraction

$$\mu = \left\langle \mathbb{P}\left[X_{[d]}\right], f\left[X_{[d]}\right]\right\rangle_{[\varnothing]} .$$

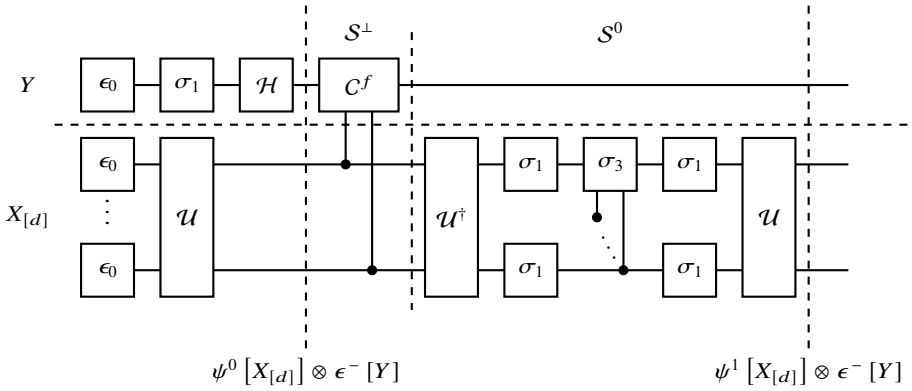


Fig. 2.18 Amplification scheme of the at the models of a Boolean function f supported coordinates of a quantum state. The ancilla variable Y is prepared as an disentagled Hadamard basis vector and stays disentangled during the following reflections. The oracle C^f effectively performs the reflection $\mathcal{S}^{\perp}$ across the bad state supported at the models of f.

We assume that we have

- a circuit $\mathcal{U}$ preparing a state with measurement distribution $\mathbb{P}\left[X_{[d]}\right]$, i.e.

$$\psi^0\left[X^{\text{out}}_{[d]}\right] = \left\langle \epsilon_{0_{[d]}}\left[X^{\text{in}}_{[d]}\right], \mathcal{U}\left[X^{\text{in}}_{[d]}, X^{\text{out}}_{[d]}\right]\right\rangle_{\left[X^{\text{out}}_{[d]}\right]}$$

  and

$$\left|\psi^0\left[X_{[d]}\right]\right|^2 = \mathbb{P}\left[X_{[d]}\right]$$

- a computation circuit for $f\left[X_{[d]}\right]$

2.4.4.1 Moments as rotation angles

We construct a Grover operator, which rotation angle determines the moment. The moment can therefore be estimated based on the measurement of the operators eigenvalue phases.

We have

$$\sum_{x_{[d]} \in [2]^d : f(x_{[d]})=1} \left|\psi^0\left[X_{[d]} = x_{[d]}\right]\right|^2 = \left\langle \mathbb{P}\left[X_{[d]}\right], f\left[X_{[d]}\right]\right\rangle_{[\varnothing]} = \mu .$$

This insight motivates us to define the orthonormal quantum states

$$\psi^{\parallel}\left[X_{[d]}\right] = \frac{1}{\sqrt{\mu}} \sum_{x_{[d]} \in [2]^d : f(x_{[d]})=1} \psi^0\left[X_{[d]} = x_{[d]}\right] \cdot \epsilon_{x_{[d]}}\left[X_{[d]}\right]$$
$$\psi^{\perp}\left[X_{[d]}\right] = \frac{1}{\sqrt{1-\mu}} \sum_{x_{[d]} \in [2]^d : f(x_{[d]})=0} \psi^0\left[X_{[d]} = x_{[d]}\right] \cdot \epsilon_{x_{[d]}}\left[X_{[d]}\right]$$

and split the prepared state into a weighted sum of them as

$$\psi^0\left[X_{[d]}\right] = \sqrt{\mu} \cdot \psi^{\parallel}\left[X_{[d]}\right] + \sqrt{1-\mu} \cdot \psi^{\perp}\left[X_{[d]}\right] .$$

Note that here the mean parameter μ is the analogon to the acceptance probability λ in quantum rejection sampling. To be more precise, when aiming to sample from the models of f using a rejection sampling scheme starting with the uniform distribution, the acceptance probability would be μ. Instead of amplifying this acceptance probability we here want to measure it to retrieve the mean parameter.

We prepare the reflection $-\mathcal{S}^0$ in the Pauli-Z scheme and the reflection $\mathcal{S}^{\perp}$ in the oracle scheme (see Sect. 2.4.3). Their concatenation results in the negated Grover operator, which causes a phase offset by π.

2.4.4.2 Retrieving the moment from the phase

We now estimate the eigenvalue phases of the Grover algorithm (see Sect. 2.4.3.3) to read of the moment.

Note that the projection of the state ψ^0 onto the eigen space to the eigen value 1 vanishes, and the corresponding trivial phase is not measured by the Quantum Phase Estimation procedure.

The phase is related to the rotation angle by (see Sect. 2.4.3.3)

$$\alpha \approx \pm \frac{1}{2\pi \mathrm{i}} \sum_{k \in [n]} \phi_k 2^{d-k-1} .$$

The rotation angle is further related to the mean μ as

$$\alpha = 2 \cdot \sin^{-1}\left(\sqrt{\mu}\right) .$$

Through measuring the phase $\phi \in \{\pm\alpha\}$ of one of these eigenvalues we can therefore retrieve the mean by

$$\mu = \left(\sin\left(\frac{\phi}{2}\right)\right)^2 .$$

2.4.4.3 Measuring the phase

We apply the Quantum Phase Estimation (Sect. 2.4.2) to measure the moment.

We prepare auxiliary qubits $A_{[p]}$ and perform for each $\ell \in [p]$ 2^ℓ repetitions of $G^{\mathbb{P},f}$ controlled on A_ℓ. The inverse Fourier transform on the auxiliary qubits is supported on the (2-adic representation of the) phases on the eigenvalues, i.e. $\pm\alpha$, and can be detected by a computational basis measure of the ancilla variables.

When measuring the phase register as $a_{[p]}$ we compute a phase

$$\phi = 2\pi \cdot \sum_{\ell \in [p]} 2^{-(\ell+1)} a_\ell$$

which is either α or $-\alpha$. We retrieve the mean in both cases by

$$\mu = \sin^2\left(\pi \cdot \sum_{\ell \in [p]} 2^{-(\ell+1)} a_\ell\right) .$$

Computational resources: Square root improvement?

Chapter 3
Quantum inference

Abstract We here investigate, how inference on Computation-Activation Networks can be executed by quantum algorithms. While we find circuits corresponding to the computation networks, we exploit Grover operators for the activation of the computed qubits. We propose two inference algorithms:

- **Quantum rejection sampling:** When the canonical parameters of an CompActNet are available, we prepare q-samples of the ancilla augmented CompActNet and apply Grover rotations to achieve an square root advantage in sampling.
- **Quantum moment matching:** When the mean parameters to an CompActNet are available, we iteratively construct the q-sample of an approximation.

3.1 Introduction

We here present quantum inference schemes for CANs. All of them exploit computation circuits as a central building block and Grover operators as an activation mechanism. While for rejection sampling we add ancilla qubits, for moment estimation and matching we do not require further qubits.

Tensor network contractions have been identified as central inference mechanism of neuro-symbolic AI. However, their exact execution is NP-hard, and one either has to restrict itself to feasible situations, or apply approximations.

Tensor Network contractions can be performed by quantum computing as follows:

- **Rejection Sampling based:** Can augment any tensor network by an Bayesian network, which is efficiently inferred by forward sampling. We prepare quantum states, which measurement statistics can be utilized to prepare samples from interesting distributions such as Computation-Activation Networks. The square root quantum advantage is obtained by amplitude amplification.
- **Phase Estimation based:** Measure the eigenvalues of a Grover operator to a formula and a q-sample.
- **Fourier Transform based:** Limited contractions (given promises on their possible values) can be formulated as Hidden Subgroup Problems and solved by Fourier transforms. We are inspired by Deutsch-Jozsa algorithm, which we generalize here to a general moment computation problem.

Note that the rejection sampling scheme and the phase estimation are ignorant under the phase of the prepared state, and only the Fourier transform based scheme depend on the phase.

Generic tensor network contraction is NP-hard, reflecting the hardness of logical and probabilistic inference. Quantum Computers can prepare states, which measurement distribution are specific network contractions, which can be hard to contract classically. We study schemes to retrieve information about these states, for example by sampling schemes, and use the retrieved information in inference.

3.1.1 Circuit Preparation schemes

Circuit preparing schemes based on approximation:

- Q-Alchemy [1], Q-Tucker [2]
- Tensor-Network Optimization based (alternating schemes) [26, 25]
- Preparing a quantum circuit given an MPS requires logarithmic depth in the tensor order [20]

Exact circuit preparing schemes for distributions:

- Grover-Rudolph [13], but no quantum advantage [16]
- Uniform controlled rotations [21]
- Quantum Shannon decomposition [30]

Circuit simulation: Since qcreason can prepare quantum circuits to arbitrary tensor networks, it can also be used to simulate quantum circuits (with an overhead!).

- [28, 27]

3.1.2 Related Works

Quantum algorithms for linear algebra:

- Contraction by SWAP and Hadamard test (see Appendix)
- Deutsch-Jozsa Algorithm
- HHL algorithm to solve linear equations [15]

3.2 Quantum Rejection Sampling

We present the Quantum Rejection Sampling method in two steps:

- **Ancilla augmentation:** Find an ancilla augmentation of a distribution, such that there is a unitary $\mathcal{U}$ preparing a state with the augmented distribution as measurement distribution
- **Acceptance amplification:** Amplify the amplitude of the ancilla qubits in state 1 by repeated application of the Grover operator

Compared with the rejection sampling approach on the start augmentation of the distribution, we achieve a square-root speed-up in the expected number of gates to be applied per sample.

We in particular apply this scheme to sample from Computation-Activation Networks, where we apply the q-sample preparation scheme of Sect. 3.2.1.

3.2.1 Preparation of q-samples to CompActNets

We investigate, how the above circuit encoding schemes can be applied in the preparation of q-samples to Computation-Activation Networks. In particular, we build compositions of computation circuits and activation circuits, for which conditional distributions coincide with Computation-Activation Networks. Note that the resulting circuits are graph-controlled circuits, equivalent to Bayesian networks.

The computation network consists already of directed cores and therefore is already a Bayesian network. The activation network however needs ancilla augmentation.

Elementary Computation-Activation Networks can be augmented by ancilla augmentation of each leg vector in the activation tensor network (see Figure 3.1). We understand ancilla variables as variables in a Bayesian Network having single parents, corresponding with the ancilla augmentation of an elementary tensor. For an example, see Figure 3.4.

To a generic activation hypergraph $\mathcal{G}$, we would do ancilla augmentation for the activation tensor network, where each hidden variable is treated as a distributed qubit. To treat it as a distributed qubit, each hidden activation qubit is prepared in uniform state (i.e. a Hadamard gate acting on a ground state) and will be omitted in the measurement scheme (either not measured, or its measurement ignored). This can be understood from a tensor network perspective,

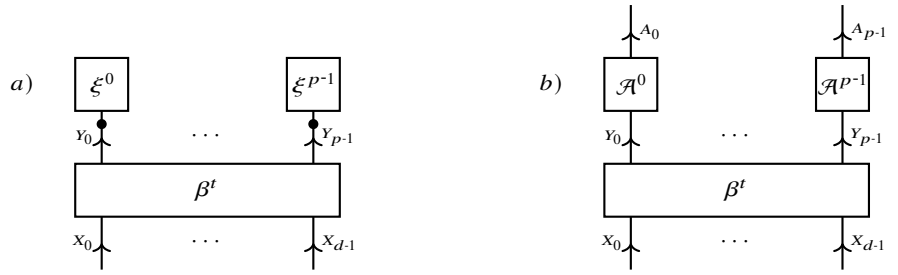


Fig. 3.1 Ancilla augmentation of an elementary Computation-Activation Network a) by replacing each leg vector of the elementary activation tensor by an directed ancilla augmentation.

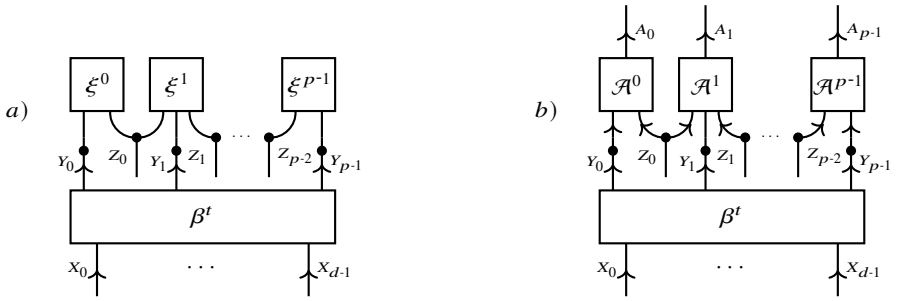


Fig. 3.2 Ancilla augmentation of an TT Computation-Activation Network a) by a Bayesian Network b). The hidden variables in the TT activation tensor are treated as distributed variables. Marginalization over the hidden variables and conditioning on the activation variables in state 1, we reproduce the Computation-Activation Network.

where marginalization means contracting, which is exactly how hidden activation variables are used. For an example in the TT format, see Figure 3.2.

3.2.1.1 Sampling from Computation-Activation Networks

For Computation-Activation Networks we find a q-sample by concatenating computation circuits and activation circuits (see Figure 3.3).

Example 3.1 (Toy Accounting Example) For a more detailed example, let us consider a system of three variables $A1$ Account 1 is booked, $A2$ Account 2 is booked, F a feature on an invoice. Assume the following two rules have to be respected:

- Exactly one account must be booked.
- If feature F is present on the invoice, the account A1 is typically booked.

We formalize this with the statistic

$$t = (X_{A1} \oplus X_{A2}, X_F \Rightarrow X_{A1}) .$$

Any elementary Computation-Activation Network with the statistic t can be realized by the circuit shown in Figure 3.4. For the preparation of the first statistic qubit Y_0 to $t_0 = X_{A1} \oplus X_{A2}$ by a computation circuit we exploit that

$$\begin{aligned} X_{A1} \oplus X_{A2} &= \epsilon_1 [X_{A1}] \otimes \epsilon_0 [X_{A2}] + \epsilon_0 [X_{A1}] \otimes \epsilon_1 [X_{A2}] \\ &= (\epsilon_1 [X_{A1}] \otimes \epsilon_0 [X_{A2}]) \oplus (\epsilon_0 [X_{A1}] \otimes \epsilon_1 [X_{A2}]) \end{aligned}$$

where by $\oplus$ we denote coordinatewise summation mod 2. The statistic qubit Y_0 is therefore prepared by two controlled NOT gates (see Figure 3.4b).

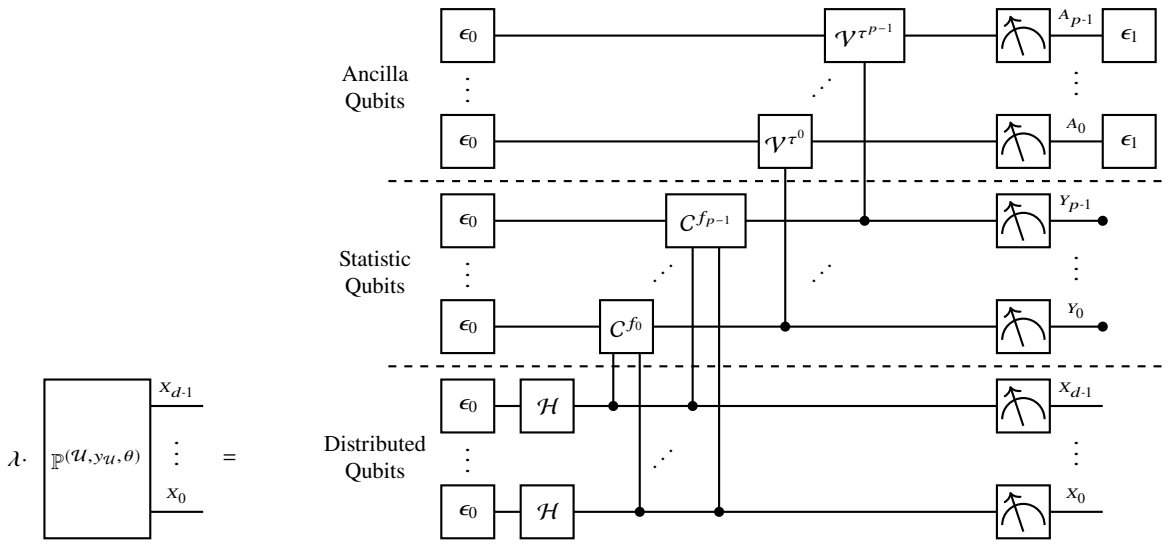


Fig. 3.3 Quantum Circuit to reproduce a q-sample of an augmented Computation-Activation Network. We measure the distributed qubits $X_{[d]}$ and the ancilla qubits $A_{[p]}$ and reject all samples, where an ancilla qubit is measured as 0.

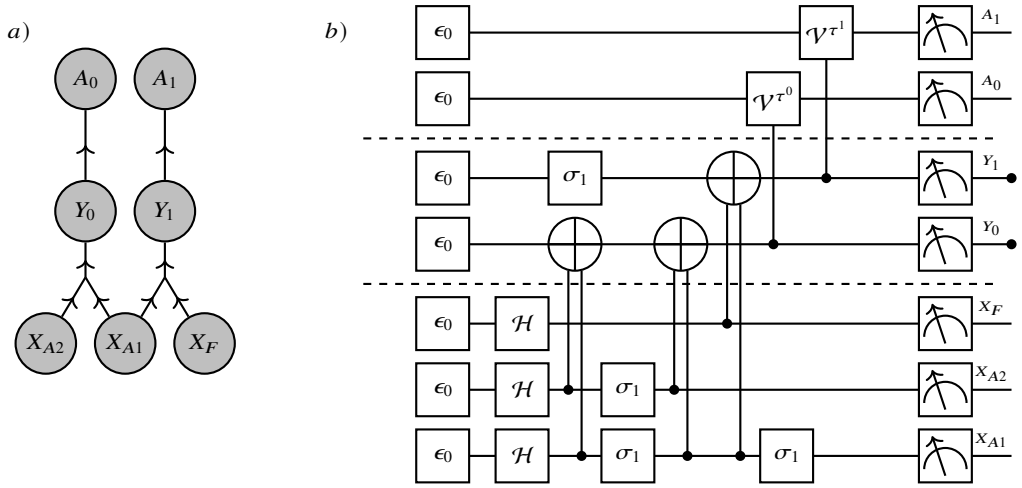


Fig. 3.4 Circuit to sample from the Computation-Activation Network in the toy accounting Example 3.1.

For the preparation of the second statistic qubit Y_1 to $t_1 = X_F \Rightarrow X_{A1}$ we exploit that the implication is True except for the case where the premise X_F is True and the head X_{A1} is False. In out mod 2 calculus, this amounts to

$$X_F \Rightarrow X_{A1} = \mathbb{I}\left[X_F, X_{A1}\right] \oplus \epsilon_1\left[X_F\right] \otimes \epsilon_0\left[X_{A1}\right] .$$

It follows that the statistic qubit Y_1 is prepared by a NOT gate (that is a Pauli-X gate σ_1) and a second controlled NOT gate (see Figure 3.4b).

For both cases, the preparation of the statistic qubit can be done by two controlled NOT gates exploiting the polynomial sparsity of the connectives.

Any non-elementary Computation-Activation Network with the statistic t can be prepared by the circuit, when choosing a single ancilla qubit, which is uniformly controlled by the qubits $Y_{[2]}$.

3.2.1.2 Acceptance Amplification

Let there be a quantum state, which measurement distribution is an ancilla augmentation $\tilde{\mathbb{P}}[A_{[p]}, X_{[d]}]$ of a target distribution $\mathbb{P}\left[X_{[d]}\right]$. We now show how Grover operators can be used to amplify the acceptance state and improve the efficiency of the rejection sampling method through an increase of the acceptance probability. This is an instance of amplitude amplification.

3.2.2 Grover construction

We relate the state $\psi^{\parallel}$ with the event $A_{[p]} = 1_{[p]}$, by constructing it as

$$\psi^{\parallel}\left[A_{[p]}, Y_{[p]}, X_{[d]}\right] = \epsilon_{1_{[p]}}\left[A_{[p]}\right] \otimes \left\langle \beta^{t}\left[Y_{[p]}, X_{[d]}\right], \sqrt{\mathbb{P}\left[X_{[d]}\right]} \right\rangle_{\left[Y_{[p]}, X_{[d]}\right]}$$

The complement state $\psi^{\perp}$ is constructed if $\lambda \neq 1$ (in which case the acceptance amplilification scheme would not be necessary)

$$\mathbb{P}^{\perp}\left[A_{[p]}, Y_{[p]}, X_{[d]}\right] = \frac{1}{1-\lambda} \cdot \left(\tilde{\mathbb{P}}[A_{[p]}, Y_{[p]}, X_{[d]}] - \lambda \cdot \epsilon_{1_{[p]}}\left[A_{[p]}\right] \otimes \mathbb{P}\left[Y_{[p]}, X_{[d]}\right]\right)$$

and

$$\psi^{\perp}\left[A_{[p]}, Y_{[p]}, X_{[d]}\right] = \sqrt{\mathbb{P}^{\perp}\left[A_{[p]}, Y_{[p]}, X_{[d]}\right]}\,.$$

We thus have

$$\psi^{0}\left[A_{[p]}, Y_{[p]}, X_{[d]}\right] = \sin\left(\frac{\alpha}{2}\right)\psi^{0}\left[A_{[p]}, Y_{[p]}, X_{[d]}\right] + \cos\left(\frac{\alpha}{2}\right)\psi^{1}\left[A_{[p]}, Y_{[p]}, X_{[d]}\right]$$

where the α is related to the acceptance probability as

$$\sin\left(\frac{\alpha}{2}\right) = \sqrt{\lambda}\,.$$

3.2.2.1 Circuit Realization

Both reflection operators are realized as their negations using the Pauli-Z scheme (see Sect. 2.4.3). The negations chancel in their concatenation, which is therefore the Grover operator.

3.2.2.2 Ancilla augmentation preserved

The measurement distribution to any quantum state in $V \backslash \mathrm{span}\,(\psi^{\perp})$ is an ancilla augmentation of $\mathbb{P}\left[X_{[d]}\right]$. When amplifying the acceptance probability we stay in the subspace V. When we do not overrotate and since we start with $\lambda > 0$ the amplified state is never in span $(\psi^{\perp})$.

3.2.3 Optimal rotation numbers

Now the application of j grover operators G on ψ^{0} prepares a state

$$\psi^{j}\left[A_{[p]}, X_{[d]}\right] = \sin\left(\left(\frac{1}{2}+j\right)\alpha\right)\psi^{\parallel}\left[A_{[p]}, X_{[d]}\right] + \cos\left(\left(\frac{1}{2}+j\right)\alpha\right)\psi^{\perp}\left[A_{[p]}, X_{[d]}\right]$$

The state is closest to $\psi^{\|}$ when

$$\left(\frac{1}{2}+j\right)\alpha \approx \frac{\pi}{2}\,.$$

Estimating for small λ (tight for small λ)

$$\sqrt{\lambda} = \sin\left(\frac{\alpha}{2}\right) \leq \frac{\alpha}{2}$$

we get a bound

$$j^{\text{opt}} \leq \frac{\pi}{4\sqrt{\lambda}}\,.$$

This shows that we have a square root improvement on classical rejection sampling, which would require $\frac{1}{\lambda}$ repetitions to produce a sample. Note however that we compare quantum circuit length with the number of classical repetitions in sampling.

3.2.4 Applications on CompActNets

The above scheme prepares the ancilla augmented Computation-Activation Network

$$\tilde{\mathbb{P}}[A_{[p]}, Y_{[p]}, X_{[d]}] = \lambda \cdot \epsilon_{1_{[p]}}\left[A_{[p]}\right] \otimes \left\langle \beta^{t}\left[Y_{[p]}, X_{[d]}\right], \mathbb{P}\left[X_{[d]}\right]\right\rangle_{\left[Y_{[p]}, X_{[d]}\right]} + (1-\lambda)\mathbb{P}^{\perp}\left[X_{[d]}, Y_{[p]}, A_{[p]}\right]$$

and the prepared state is the combination of q-samples

$$\psi^{0}\left[A_{[p]}, Y_{[p]}, X_{[d]}\right] = \sqrt{\lambda} \cdot \epsilon_{1_{[p]}}\left[A_{[p]}\right] \otimes \left\langle \beta^{t}\left[Y_{[p]}, X_{[d]}\right], \sqrt{\mathbb{P}\left[X_{[d]}\right]}\right\rangle_{\left[Y_{[p]}, X_{[d]}\right]} + \sqrt{(1-\lambda)}\sqrt{\mathbb{P}^{\perp}\left[X_{[d]}, Y_{[p]}, A_{[p]}\right]}\,.$$

Sampling from the augmented Computation-Activation Network can be also done classically by forward sampling: Just draw the variables from uniform, calculate the value qubit by a logical circuit inference and accept with probability by the computed value. To achieve a quantum advantage we manipulate the circuit first by the amplitude amplification scheme and increase the acceptance probability.

3.2.5 Examples

Example 3.2 (Sample a model of a propositional formula)

Consider the task of sampling from the models of a propositional formula $f\left[X_{[d]}\right]$, which can be expressed as a Computation-Activation Network as

$$\frac{1}{\langle f\rangle_{[\varnothing]}} \cdot f\left[X_{[d]}\right] = \frac{1}{\langle f\rangle_{[\varnothing]}} \cdot \left\langle \beta^{f}\left[Y, X_{[d]}\right], \epsilon_{1}\left[Y\right]\right\rangle_{\left[X_{[d]}\right]}\,.$$

Note that in this case, $\frac{1}{2^{d}} \cdot \beta^{f}\left[Y, X_{[d]}\right]$ is an ancilla augmentation of $\frac{1}{\langle f\rangle_{[\varnothing]}} \cdot f\left[X_{[d]}\right]$ with Y as the ancilla variable, which q-sample is prepared by the oracle C^{f} acting on the ground state $\epsilon_{0_{[d+1]}}\left[Y, X_{[d]}\right]$. The acceptance rate is then

$$\lambda = \frac{1}{2^{d}} \cdot \left\langle \beta^{f}\left[Y=1, X_{[d]}\right]\right\rangle_{[\varnothing]} = \frac{\langle f\rangle_{[\varnothing]}}{2^{d}}$$

and the optimal number of rotations is

$$j^{\text{opt}} = 2 \cdot \arcsin\left(\sqrt{\frac{\langle f \rangle_{[\varnothing]}}{2^d}}\right).$$

The example of a single rotation is sketched in Figure 3.5.

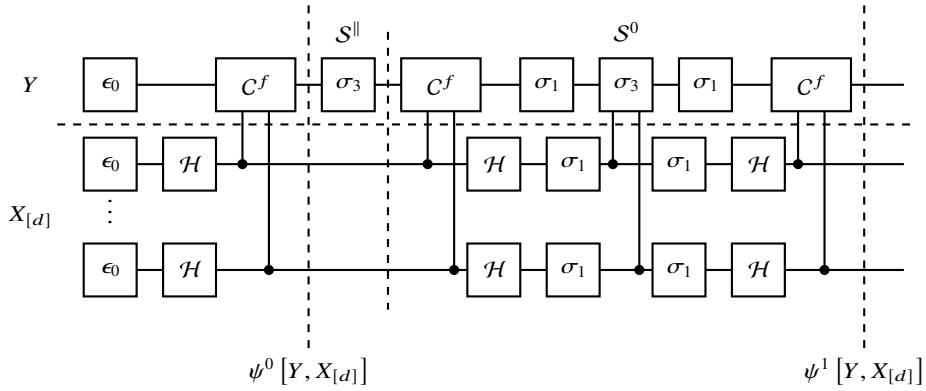


Fig. 3.5 Amplitude amplifying circuit for the ancilla augmentation of a function f, for $j = 1$ repetitions. The initial state $\psi^0\left[Y, X_{[d]}\right]$ is the q-sample of the by Y augmented uniform distributions among the models of f, and prepared by action of the oracle C^f on the ground state. The amplification is then done by the Grover operator, which consists in the reflection $\mathcal{S}^{\|}$ across the state $\psi^{\|}\left[Y, X_{[d]}\right]$ followed by a reflection reflection $\mathcal{S}^0$ across the state $\psi^0\left[Y, X_{[d]}\right]$.

Example 3.3 (Sampling from the toy accounting system) For the toy accounting system we have

$$\max_{x_F, x_{A1}, x_{A2} \in [2]} \mathbb{P}\left[X_F = x_F, X_{A1} = x_{A1}, X_{A2} = x_{A2}\right] = \begin{cases} \frac{1}{3+\exp[-\theta]} & \text{if} \quad \theta \geq 0 \\ \frac{1}{3 \cdot \exp[\theta]+1} & \text{if} \quad \theta < 0 \end{cases}$$

and therefore the acceptance probability when sampling (X_F, X_{A1}, X_{A2}) from uniform by

$$\lambda = \frac{1}{8} \cdot \left(\max_{x_F, x_{A1}, x_{A2} \in [2]} \mathbb{P}\left[X_F = x_F, X_{A1} = x_{A1}, X_{A2} = x_{A2}\right]\right)^{-1} = \begin{cases} \frac{3+\exp[-\theta]}{8} & \text{if} \quad \theta \geq 0 \\ \frac{3 \cdot \exp[\theta]+1}{8} & \text{if} \quad \theta < 0 \end{cases}.$$

The optimal number of amplifications is then

$$j^{\text{opt}} = \begin{cases} 0 & \text{if} \quad \theta \geq \theta_{\text{critical}} \\ 1 & \text{if} \quad \theta < \theta_{\text{critical}} \end{cases}$$

for

$$\theta_{\text{critical}} = \ln\left[1 - \frac{4\cos(1)}{3}\right] \approx -1.27\,.$$

Note that in the limit of $\theta \to -\infty$, we effectively have a single model and therefore a minimum of the acceptance probability among the distributions of three Boolean variables.

3.3 Moment Amplification

Instead of measuring the phases of the effective Grover operator, we now investigate the effect of uncontrolled Grover operations on the distributions. We find an analogon in discrete increases of the canonical parameters when the measurement distribution of the start state is in an exponential family.

3.3.1 Interpretation of Grover rotations

Rotating j times across the initial state then gives the tensor product of the antisymmetric ancilla state with

$$\psi^j\left[X_{[d]}\right] = \sin\left(\left(\frac{1}{2}+j\right)\alpha\right)\psi^{\mathbb{P}^0,f}\left[X_{[d]}\right] + \cos\left(\left(\frac{1}{2}+j\right)\alpha\right)\psi^{\mathbb{P}^0,\neg f}\left[X_{[d]}\right] .$$

This reflection can be performed by the operator on the distributed qubits

$$U^t\left(\delta\left[X^{\text{in}}_{[d]}, X^{\text{out}}_{[d]}\right] - 2\epsilon_0\left[X^{\text{in}}_{[d]}\right]\otimes\epsilon_0\left[X^{\text{out}}_{[d]}\right]\right)U .$$

3.3.1.1 Mean parameter interpretation

The mean parameter of f with respect to the measurement distribution is

$$\mu^j = \left\langle \mathbb{P}^j\left[X_{[d]}\right], f\left[X_{[d]}\right]\right\rangle_{[\varnothing]} = \sin\left(\left(\frac{1}{2}+j\right)\alpha\right)^2 = \sin\left((1+2j)\sin^{-1}\left(\sqrt{\mu^0}\right)\right)^2$$

Here we used that $\alpha = 2\cdot\sin^{-1}\left(\sqrt{\mu^0}\right)$.

3.3.1.2 Canonical parameter interpretation

Lemma 3.1 *Works also for more generic measurement distributions! Let ψ^0 be the q-sample of the elementary Computation-Activation Network with canonical parameters $(\mathcal{U}, y_{\mathcal{U}}, \theta)$, and let us amplify the formula f_ℓ. Then ψ^j is the q-sample of the elementary Computation-Activation Network with canonical parameters*

$$\mathcal{U}^j = \begin{cases} \mathcal{U}\cup\{\ell\}, y^\ell_{\mathcal{U}} = (y_{\mathcal{U}}, 1) & \text{if} \quad \sin\left((1+2\cdot j)\sin^{-1}\left(\sqrt{\mu_0\left[L=\ell\right]}\right)\right) = 1 \\ \mathcal{U}, y_{\mathcal{U}} & \text{else} \end{cases}$$

and

$$\theta_j\left[L=\ell\right] = \begin{cases} 0 & \text{if} \quad \sin\left((1+2\cdot j)\sin^{-1}\left(\sqrt{\mu_0\left[L=\ell\right]}\right)\right) = 1 \\ \theta\left[L=\ell\right] + \frac{\cos^2\left(\frac{\alpha}{2}\right)\left(1-\cos^2\left(\left(\frac{1}{2}+j\right)\alpha\right)\right)}{\sin^2\left(\frac{\alpha}{2}\right)\cdot\cos^2\left(\left(\frac{1}{2}+j\right)\alpha\right)} & \text{else} \end{cases} .$$

Proof. Notice, that

$$\left\langle \mathbb{P}^0\left[X_{[d]}\right], \beta^{f_\ell}\left[Y_\ell, X_{[d]}\right]\right\rangle_{[Y_\ell]} = \begin{pmatrix}\left(\cos\left(\frac{\alpha}{2}\right)\right)^2 \\ \left(\sin\left(\frac{\alpha}{2}\right)\right)^2\end{pmatrix}[Y_\ell] = \begin{pmatrix}1-\mu_0\left[L=\ell\right] \\ \mu_0\left[L=\ell\right]\end{pmatrix}[Y_\ell]$$

and

$$\left\langle \mathbb{P}^j\left[X_{[d]}\right], \beta^{f_\ell}\left[Y_\ell, X_{[d]}\right]\right\rangle_{[Y_\ell]} = \begin{pmatrix} \left(\cos\left(\left(\frac{1}{2}+j\right)\alpha\right)\right)^2 \\ \left(\sin\left(\left(\frac{1}{2}+j\right)\alpha\right)\right)^2 \end{pmatrix} [Y_\ell] \begin{pmatrix} 1-\mu_j\left[L=\ell\right] \\ \mu_j\left[L=\ell\right] \end{pmatrix} [Y_\ell]$$

If $\mu_j\left[L=\ell\right]=1$ (that is $\sin\left(\left(\frac{1}{2}+j\right)\alpha\right)=1$), the amplitude amplification amounts to adding f_ℓ as a hard constraint. This is equal to $\mathcal{U}^j=\mathcal{U}\cup\{\ell\}$ and $y^j_{\mathcal{U}}=(y_{\mathcal{U}},1)$.

If $\mu_j\left[L=\ell\right]\neq 1$ we choose $\theta_\Delta\left[L=\ell\right]\geq 0$ as in the claim and have after some algebra

$$\frac{\left(\cos\left(\frac{\alpha}{2}\right)\right)^2}{\left(\cos\left(\frac{\alpha}{2}\right)\right)^2+\theta_\Delta\left[L=\ell\right]\cdot\left(\sin\left(\frac{\alpha}{2}\right)\right)^2} = \left(\cos\left(\left(\frac{1}{2}+j\right)\alpha\right)\right)^2$$

and therefore

$$\left\langle \mathbb{P}^j\left[X_{[d]}\right], \beta^{f_\ell}\left[Y_\ell, X_{[d]}\right]\right\rangle_{[Y_\ell]} = \left\langle \exp\left[\theta_\Delta\left[L=\ell\right]\cdot f_\ell\left[\theta\right]\right]\mathbb{P}^0\left[X_{[d]}\right], \beta^{f_\ell}\left[Y_\ell, X_{[d]}\right]\right\rangle_{[Y_\ell|\varnothing]}.$$

Thus, the rotation of the q-sample corresponds with an increase of the canonical parameter $\theta_0\left[L=\ell\right]$ by $\theta_\Delta\left[L=\ell\right]$. □

3.3.1.3 Maximum Entropy

When $\left(\frac{1}{2}+j\right)\alpha\leq\frac{\pi}{2}$ then all amplitudes remain positive, and $\psi^j\left[X_{[d]}\right]$ is a q-sample. What is more, we can show that the property of maximum entropy with respect to a statistic containing the amplified formula.

Theorem 3.1 *Let ψ^0 be the q-sample of a maximum entropy distribution with respect to the uniform base measure and the statistic $\mathcal{F}$ containing the amplified formula, and let $j\in\mathbb{N}$ be a rotation number such that $\left(\frac{1}{2}+j\right)\alpha\leq\frac{\pi}{2}$. Then also ψ^j is the q-sample of a maximum entropy distribution.*

Proof. By Lem. 3.1 the amplitude amplified state is an elementary Computation-Activation Network and thus a maximum entropy distribution. □

Notice, that we can only increase the canonical parameter by amplitude amplification. Decreasing could be done by amplitude amplification on $\neg f_\ell$ instead.

3.3.2 Moment Matching

We now investigate which Computation-Activation Networks can be prepared directly with amplitude amplification. While we already have studied sparse representations of the computation network parts, we here apply amplitude amplification by Grover rotations as an activation mechanism.

We orient on the iterative proportional fitting algorithm, which is used to estimate the canonical parameters in exponential families [31] and more generally of Computation-Activation Networks [10].

The algorithm loops over the moment estimation and moment amplification subroutines, which have been introduced in the sections above. The idea is to first compute the current moments of a distribution and second amplify one.

We understand this rotation as a moment matching operation to the formula f, see Algorithm 1.

We can use Algorithm 1 as a preparation algorithm of a q-sample before doing ancilla augmentation as in the main section. Note that is important to keep track of the canonical parameters in the preparation. When activating the statistic qubits, only the difference of the already prepared canonical parameters and the target canonical parameters has to be used (importance sampling interpretation).

Algorithm 1 Quantum Circuit Moment Matcher

Require: Formulas $f_p\left[X_{[d]}\right]$ for $\ell \in [p]$, vector $\mu_D[L]$ of mean parameters
Ensure: Circuit $\mathcal{U}$ preparing a state (when applied on the ground state) which measurement distribution matched $\mu_D[L]$

$U \leftarrow (\sigma_1 \circ \mathcal{H})[A_{\text{in}}, A_{\text{out}}] \otimes \left(\bigotimes_{k\in[d]} \mathcal{H}\left[X_{k,\text{in}}, X_{k,\text{out}}\right]\right)$
while Convergence criterion not met **do**
Estimate $\mu[L]$,

- either by particle bases inference
- or by quantum counting (measuring the eigenvalue of the Grover operator), see Sect. 2.4.4

Select an $\ell \in [p]$, which moment has to be updated (e.g. by comparison of $\mu[L]$ with $\mu_D[L]$)
Choose whether to amplify f_ℓ or $\neg f_\ell$ based on whether the estimated mean is smaller than the target mean
Compute optimal j to match $\mu_D[L]$
Extend Circuit $\mathcal{U}$ by j alternations of

- C^{f_ℓ} (effectively an reflection across the subspace spanned by one-hot encoded models)
- reflections across the current state (by $U(I - 2e_0e_0)U$)

end while

3.4 Extension: Sampling from proposal distributions

We can prepare basis circuit encodings to selection augmented formulas, in this way introducing formula selecting networks.

Idea for an inductive reasoning scheme: Prepare a q-sample from the empirical distribution and the current distribution. Then prepare the basis circuit encodings, where the selection variables are shared and the distributed variables assigned to the prepared samples. Now, the ancilla qubits can be designed to ϵ_1 and ϵ_0 accordingly. The rejection sampling scheme on both ancillas being 1 and the measurement of L prepares then the distribution

$$\left\langle \left\langle \mathbb{P}^D\left[X_{[d]}\right], \sigma^t\left[X_{[d]}, L\right]\right\rangle_{[L]}, \left\langle \tilde{\mathbb{P}}\left[X_{[d]}\right]\right\rangle_{[L]}\right\rangle_{[L|\varnothing]}$$

That is, the probability of selecting ℓ is proportional to

$$\mu_D[L = \ell] \cdot (1 - \tilde{\mu}[L = \ell])$$

and thus prefers formulas, which have a large empirical mean, but a small current mean.

Open Question: Since the distribution is "similar" to $\exp[\mu_D[L = \ell] - \tilde{\mu}[L = \ell])]$ (terms appear in Taylor of first order), can we tune the distribution with an inverse temperature parameter β?

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